

Research map

Interesting mathematics, faithful entrances,
and the questions still ahead.

63	231	90
field dossiers	surveyed question seeds	facilitator planning cards

First breadth edition / map v1
25 September 2026

This volume preserves all three research surveys and their source locators. The companion volume contains the 90 reviewed planning records. These counts describe different layers of the library; they are not counts of elementary sessions or completed research subjects.

Companion volume: [Mathematics atlas - investigation plans](#)

Prepared for the Lowell Math Circle library in Bellingham. Capability prerequisites replace fixed grade placement. No new atlas activity is represented as classroom-piloted.

Contents

Field headings are linked. Use PDF bookmarks to move directly between the map, the 63 dossiers, source notes, and the frontier. Each seed has a stable field-based ID; planning-card IDs are shown beside their primary field.

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Reading this map

The atlas begins with an adult mathematical question and searches for a faithful investigation at an explicit prerequisite level. A finite special case can be real mathematics without exhausting its field. Some worthwhile entrances require complex numbers, matrices, calculus, or university-level definitions; those gates remain visible.

The 63 dossiers are a complete top-level breadth pass through the pinned MSC snapshot. Their 231 numbered seeds are provisional research proposals, not a de-duplicated count of mechanisms. The companion 90 cards are a selected and expanded planning library: 24 algebra/discrete, 36 geometry/analysis, and 30 probability/application records. They represent all 60 substantive top-level fields as primary topics. Fields 00, 01, and 97 have supporting dossiers.

The surveys are retained as screening records. References in them to future family design or to work not yet completed describe that research stage. Family-level mathematical/editorial review and repairs have since been recorded separately. Agent review does not establish classroom suitability, and none of the new atlas families has classroom evidence.

All 534 subareas, 5,503 ordinary subject entries, and 503 cross-cutting entries remain individually unvisited. Top-level representation gives them no inherited coverage. A source contents page supports scope; inspected relevant text or a self-contained proof is needed to support a particular theorem. Failed retrievals, excerpt-only access, and older editions remain visible in the dossiers.

The low entrances are proposals, not promises about age. Reading may be supplied orally; arithmetic, spatial tracking, abstraction, and proof may develop independently. A satisfying early stop need not reach the most advanced theorem. For an actual meeting, use the companion planning card rather than treating a research seed as a finished session.

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Layer	Extent	Meaning
Top-level fields	63 dossiers	Screened; 60 primary content fields plus 3 support fields
Research seeds	231	Full seed text printed in its field dossier
Investigation plans	90	Reviewed planning records in the companion book
Taxonomy descendants	6,540 rows	All individually unvisited; no inherited coverage
Classroom pilots	0	No claimed evidence of use for new atlas designs

Map lock and coverage contract

The first breadth survey is locked on 25 September 2026 as **atlas map v1**. Lock means stable field ownership and IDs for the next design pass, not mathematical infallibility or a claim that all interesting problems have been found. Review may correct claims and future editions may add anchors.

All 63 top-level fields in the pinned official MSC snapshot have one substantive dossier: 16 in algebra/discrete/foundations, 27 in geometry/topology/analysis, and 20 in probability/computation/applications. Fields 00, 01 and 97 have explicit cross-cutting research/support dispositions. The other 60 receive student-investigation development, at the actual prerequisite level the mathematics warrants.

The coordinator read the three surveys and reconciled headings against the imported code list. The independent undergraduate-syllabus check is in reviews/coverage-audit.md. The result supports proceeding with the breadth design pass. It does **not** support marking any of the 534 subareas or 6,006 five-character entries complete. In particular, a source's table of contents establishes scope; only inspected relevant text or a supplied self-contained proof supports the specific theorem used in a family.

Design assignments

Volume	Family IDs	Target	Requirement
Algebra and discrete mathematics	AD-01 onward	24 families	All 14 substantive fields; protect noncommutative, homological and K-theoretic gates.
Geometry and analysis	GA-01 onward	36 families	All 27 fields; genuine continuous questions and honest SCV/global-analysis stops.
Probability and applications	AP-01 onward	30 families	All 19 substantive fields; distinct inference, physical models, control and strategic games.

The 90-family target is an editorial breadth allocation, not a count of distinct mathematical disciplines or a promise of 90 elementary sessions. Each record follows family-format.json. A direct-source check and the exact example are author obligations; independent review follows. A family may use a shared mechanism as long as its limitation is explicit. Do not turn a difficult topic into a falsely exact simple analogy to fill a quota.

Priorities beyond field counts: learner choices, surprise, a question that rewards explanation, low-preparation objects, and extensions that change the mathematics. The existing Weeks 1-10 remain a quality benchmark and duplicate-check corpus. Their topic labels are not reused as credit for new work.

PART I

Algebra, discrete mathematics, and foundations

Stage: breadth-first research dossier, 25 September 2026. **Assigned MSC2020 fields:** 00, 01, 03, 05, 06, 08, 11-20: sixteen top-level fields. This is a research map and a candidate inventory, not a set of reviewed activities, a complete subfield census, or classroom evidence. The original atlas draft is advisory. The questions below are original design proposals unless a source exercise is explicitly identified.

Read against README.md, math_circle_atlas_plan.md, and plans/fall-weeks-02-10-mathematical-redesign.md. Current Weeks 1-10 are planned material, not documented successful pilots. They already occupy a substantial part of the elementary discrete entrance; a new story around toggles, permutations, or Euler routes must not be counted as new mathematics.

How to interpret the entrances

E - exact: the proposed finite or restricted problem is genuinely an instance of the stated structure. **R - representation:** the objects and actions faithfully realize that structure. **A - analogue:** a useful shared mechanism that does not establish the advanced result. **G - gated:** meaningful work remains at the stated prerequisite level. Every entrance still needs a separate design-and-review pass.

The capability gates are not grades. **Objects** means sorting, matching, spatial manipulation, and following a demonstrated rule; reading may be supplied orally. **Cases** adds systematic recording, comparison, and reasons covering every case. **Arithmetic** adds the operations explicitly named. **Symbols** adds equations, variables, and functions. **Linear** means vectors, matrices, and linear maps; **Structural** means sets of objects/maps, quotients, proof, and universal properties. These are independent capabilities, not a compulsory ladder.

FIELD 00 | 3 SURVEYED SEEDS | support dossier

00 General mathematics: a cross-cutting facet

This category cannot be honestly “covered” by one puzzle. It includes exposition, philosophy, problem solving, and general mathematical work. Preserve it as a facet and use it to ask what kind of mathematical claim an investigation produces. The official MSC is a research-literature taxonomy, not a list of classroom subjects [S0].

Three productive questions run across the atlas:

00-S01 What would convince another person? Offer a construction, an impossibility claim, and an optimality claim on the same small board. For example, a route demonstrates existence; a parity obstruction rules out a route; a route matching a lower bound proves optimality. Ask which claims a photograph certifies and which need a reason. **E:** this is actual mathematical justification, with an Objects entrance and a Cases explanation gate.

00-S02 What has been forgotten in a model? Redraw a network while preserving endpoints of every edge. Does the route question change? Then merge two vertices and ask again. Children decide which changes preserve the problem, instead of merely being told that the picture is a graph. **E/R:** graph isomorphism and a counterexample to an invalid quotient. Week 10 is a direct prior instance; the new question would be model equivalence, not another route puzzle.

00-S03 Which conjecture deserves to survive? Generate examples, seek a counterexample, repair the hypothesis, then distinguish the repaired statement from its converse. Euclid's finite-list prime argument is a useful adult calibration: “the constructed number is prime” is stronger than, and unnecessary for, the proof [S2]. A child who understands factors can find a counterexample to the stronger claim.

Stop/trap: these are proof-and-model facets, not substitutes for foundations, philosophy of mathematics, or mathematical education research. A checklist that says “uses reasoning” adds no substantive coverage. The next survey must separately investigate general mathematical modeling and philosophy; this dossier does not settle either.

FIELD 01 | 3 SURVEYED SEEDS | support dossier

01 History and biography: questions with provenance

History concerns how mathematics actually developed, including notation, tools, institutions, and people. Do not use legendary anecdotes as evidence. Three source-grounded mathematical comparisons are available now:

01-S01 Why does repeated subtraction find a common measure? Reconstruct Euclid VII.2 with two lengths made of unit blocks. Compare subtracting one copy at a time with removing several copies. **E**: the invariant common-divisor set and termination are visible; division notation is optional [S1].

01-S02 Does a finite list of primes exhaust the possibilities? Reconstruct IX.20 and examine why the new number need not be prime. **E**: a new prime divisor, not necessarily a new prime equal to the constructed integer, defeats a proposed exhaustive list [S2]. The child gate is multiplication, divisibility, and the meaning of “any proposed list.”

01-S03 What did Euler remove from the bridge map? Compare a geographic drawing, a sequence of region letters, and a modern graph. Which information survives? The Euler Archive identifies the original work E53 and distinguishes written and publication dates [S3]. **E**: the route problem; **A**: claiming the modern graph formalism exactly reproduces Euler's presentation without reading that presentation.

These should become comparative investigations, not mini-biographies preceding arithmetic. At a later gate, ask how a proof changes when zero is admitted as a number, or when algebraic notation replaces geometric magnitudes; those require further historical sources.

Stop/trap: three European-source anchors do not cover mathematical history. Egyptian, Babylonian, Chinese, Indian, Islamic, Indigenous, and modern global mathematical traditions remain an explicit research frontier, with authored historical scholarship needed. “Ancient people did our modern worksheet” is an anachronism. Existing Weeks 4/9 overlap Euclidean divisibility and Week 10 overlaps Euler; historical framing alone does not create new mechanism coverage.

FIELD 03 | 4 SURVEYED SEEDS | AD-01; AD-02

03 Mathematical logic and foundations

Distinguishing territory includes truth versus proof, expressibility, models, computation, set-theoretic size, and independence. A collection of truth puzzles alone misses most of it. Open Logic supplies inspected anchors for diagonalization, first-order compactness, and undecidability [S4].

03-S01 Can two different rule cards accept exactly the same objects? On a finite inventory, compare “red and round” with the negation of “not red or not round”; then alter a quantifier. **E:** Boolean semantics and equivalence. Objects suffice for participation; Cases is needed to justify equivalence over all possible property combinations.

03-S02 Can you defeat every row in a proposed list? Given an $(n \times n)$ binary array, construct a length- (n) row differing from row (i) at position (i) . **E:** finite diagonal construction. The infinite binary-sequence argument needs a separately stated rule for every positive integer. It proves no enumeration of all infinite binary sequences exists; the finite activity alone does not prove uncountability.

03-S03 Can every finite set of demands be met, while the whole collection requires a new kind of model? Explore demands “there are at least (n) objects,” then the infinite set of all such demands. **E:** concrete finite models; **G:** first-order compactness requires syntax, satisfaction, and quantifiers. Do not claim that every infinite puzzle inherits compactness under arbitrary rules.

03-S04 Can a machine reliably predict whether all other machines stop? First trace tiny explicitly specified programs, distinguishing a terminating computation from an observed prefix. **E:** computations; **G:** the diagonal impossibility argument requires encodings and self-application. A looping toy machine is not a proof of undecidability.

Frontier: proof theory, constructive logic, forcing, large cardinals, descriptive set theory, and reverse mathematics remain unscreened internally. Week 5 supplies finite constraint satisfaction and Week 6 elimination of candidates; neither already covers model theory or incompleteness. Useful early designs are truth-table equivalence and diagonal adversarial construction; advanced claims must remain gated.

FIELD 05 | 4 SURVEYED SEEDS | AD-03; AD-04; AD-05

05 Combinatorics

Counting, extremal structure, designs, graph theory, algebraic combinatorics, and random constructions have different mathematical questions. Four precise anchors from Stanley are available [S5]:

05-S01 How many walks of a fixed length reach a destination? Children move counters on a small network and count by last step. The (i,j) entry of (A^k) counts length- (k) walks for the specified adjacency convention. **E:** recurrence on a finite graph; matrix multiplication can arrive after the combinatorial reasoning. Distinguish walks from simple paths.

05-S02 How many subsets can be chosen if none may contain another? Begin with all subsets of four named objects. Children choose a large incomparable collection and challenge another group's certificate. **E:** antichains in a Boolean lattice. The general largest size is $\text{choose}(n, \text{floor}(n/2))$; full proof needs a chain or counting argument, not just inspection of the middle layer.

05-S03 How many ways can a network stay connected after deleting all unnecessary links? Enumerate spanning trees of a square with one diagonal. **E:** tree enumeration. The Matrix-Tree Theorem connects the count to a Laplacian cofactor for a finite connected loopless graph; determinant evaluation is a later gate.

05-S04 Can every labeled tree be encoded by a short list and decoded uniquely? Try leaf deletion on four labels before deriving Prüfer coding. **E:** bijection; for $(n \geq 2)$, the eventual count (n^{n-2}) is a theorem about labeled trees, not unlabeled shapes.

Access: Objects for constructing, Cases for exhaustive lists and bijections, arithmetic for counts, Linear for spectral extensions. **Stop/trap:** a huge brute-force list is not automatically an engaging investigation; proving completeness must remain possible at the chosen scale. Source-dependent graph conventions matter, particularly loops.

Frontier: extremal/Ramsey theory, probabilistic method, matroids, design existence, enumeration via generating functions, analytic combinatorics, and additive combinatorics need independent dossiers. Weeks 1/5/7/8/10 already cover tilings, Latin constraints, impartial games, and routes. Antichains and tree codes supply genuinely different questions.

FIELD 06 | 4 SURVEYED SEEDS | AD-06; AD-07

06 Order, lattices, ordered algebraic structures

The defining move is replacing a single line of “bigger/smaller” with partial comparison and asking about common bounds. Burris-Sankappanavar I §§1-5 distinguishes lattices, distributivity, complete lattices, and closure [S6].

06-S01 Can two collections be incomparable? Order subsets by inclusion, or positive divisors of 12 by divisibility. Find a greatest common lower bound and least common upper bound. **E/R:** meet/join, respectively intersection/union or gcd/lcm. Objects can handle subset cards; divisibility requires multiplication or supplied division diagrams.

06-S02 Does distributing one operation across another always work? Compare subset lattices with the five-element diamond (M_3). With distinct middle elements (a,b,c), test $(a \wedge (b \vee c))$ against $((a \wedge b) \vee (a \wedge c))$. **E:** a small exact counterexample; a Hasse diagram is a new representation to teach, not a prerequisite to assume.

06-S03 What is forced by a chosen starting collection? On a four-object system with explicit implications, repeatedly add forced objects. Compare all legal orders of applying rules. **E:** extensive, monotone, idempotent closure. Ask whether closing the union differs from merely uniting two closed sets; the closed sets form a lattice whose join may require closing again.

06-S04 How can two comparison systems translate optimally into one another? “How many three-seat benches are needed?” and “How many children can these benches seat?” instantiate $\text{ceil}(x/3) \leq n$ if and only if $x \leq 3n$, for nonnegative integer counts. **E:** a Galois connection of preorders, with rounding/division gate [S7, §1.4].

Stop/trap: not every poset is a lattice, not every lattice is distributive, and a closure procedure does not establish all of lattice theory. Ordered groups, vector lattices, and Boolean duality are an advanced frontier. Week 1's tiling-order ideas are related but do not license a blanket lattice label.

FIELD 08 | 4 SURVEYED SEEDS | AD-08

08 General algebraic systems / universal algebra

This field studies operations and identities across many kinds of algebra, including when quotienting, combining, or freely generating structures preserves laws. Its interest is in the common structure, not merely using unusual arithmetic [S6, II §§1-11; III §§1-4].

08-S01 Can a binary-operation table obey some laws and violate others? Provide three symbols and let children invent a complete table; challenge commutativity, associativity, or an identity with explicit witnesses. **E:** finite algebras. A small table can have a substantial question; asking children to verify all triples mechanically cannot be the whole activity.

08-S02 Which symbols may safely be treated as the same? Start with addition modulo four. Merge 0 with 2 and 1 with 3; compare with merging only 0 with 1. **E:** the first partition is a congruence, the second fails well-definedness. Children must test whether choices of representative change an answer.

08-S03 Can two machines run side by side and retain the same laws? Build pairs of states and apply each operation coordinatewise. **E:** direct product. The interesting question is what properties survive and whether every paired state can actually be reached from selected generators; do not assume all products are cyclic.

08-S04 What is the most general machine with two buttons and no extra rules? Use words in two letters; concatenate, then optionally impose $(aa=a)$, or $(ab=ba)$, and investigate equivalent words. **E:** free monoids and presentations. **G:** Birkhoff's HSP theorem concerns equational classes of a fixed signature and closure under homomorphic images, subalgebras, and products; word play is not its proof.

Stop/trap: universal algebra and algebraic geometry use “variety” differently. A Latin square is a quasigroup table but need not be a group table. Week 5 is a candidate exact quasigroup connection, yet tower-clue uniqueness is a different question from identity preservation. Constraint-satisfaction polymorphisms and decidability remain research frontier.

FIELD 11 | 4 SURVEYED SEEDS | AD-09; AD-10

11 Number theory

Arithmetic patterns become serious mathematics through divisibility, local obstructions, global solutions, approximation, and arithmetic structure. Four different mechanisms should be kept apart:

11-S01 Can one hidden number meet two remainder reports? Use two rotating rings of sizes 3 and 4, then 4 and 6. For coprime positive moduli (m,n) , every residue pair has exactly one solution modulo (mn) [S8, Theorem 1.1]. **E:** Chinese remaindering. For the changed pair, children must discover incompatibility instead of being told the same theorem applies.

11-S02 Can a number be both triangular and square? Build 1 and 36 with dots, then seek the next example. The transformation $(m(m+1)/2=n^2 \Leftrightarrow (2m+1)^2-2(2n)^2=1)$ connects the question exactly to a restricted Pell equation [S9, §3]. **E:** the dot question; **G:** proving all solutions requires more than extending a numerical pattern. The recurrence from multiplying by $(3+2\sqrt{2})$ can be checked symbolically once quadratic expressions are available.

11-S03 Can a local test rule out a global integer solution? Ask whether $(x^2+y^2=3 \pmod{4})$ can occur: squares have residues 0 or 1, so it cannot. **E:** a modular obstruction. Then ask whether passing one test guarantees a solution; it does not. This is an original small instance, not a claim to have taught the arithmetic local-to-global principle.

11-S04 Can you always escape a finite proposed prime list? Euclid's argument gives a complete elementary impossibility proof [S2]. Change “prime” to other multiplicatively defined lists only after checking which argument survives.

Access: remainders can be tracked by movement before division algorithms; Pell begins concretely but its classification stops at Symbols with an additional proof gate. **Frontier:** analytic prime distribution and (L) -functions, algebraic number fields, (p) -adic arithmetic, elliptic curves, transcendence, Diophantine approximation, and arithmetic dynamics remain substantive future branches. Weeks 4 and 9 cover gcd/orbits; they are not new coverage here.

FIELD 12 | 3 SURVEYED SEEDS | AD-11; AD-12

12 Field theory and polynomials

The central questions concern which equations become solvable after enlarging the number system, how roots can be permuted while preserving arithmetic, and what new operations permit. Three distinct bridge candidates:

12-S01 Can arithmetic with four symbols allow division by every nonzero symbol? Construct $F_2[t]/(t^2+t+1)$, representing $(0, 1, t, 1+t)$ with two-bit cards. Compare with arithmetic modulo four. **E:** the former is a field; the latter has zero divisors. The irreducible-polynomial requirement matters [S10, Theorem 1.1 and warning]. Objects plus table-following permits exploration; polynomial reduction and proof are later gates.

12-S02 Which changes to roots preserve every arithmetic relation? In $\mathbb{Q}(\sqrt{2})$, test $(a+b\sqrt{2} \mapsto a-b\sqrt{2})$. Then compare two independent square roots. **E:** automorphisms of a specified extension, not arbitrary permutations of a printed list of roots. The Symbols gate includes fractions, distributivity, and the distinction between a number and its decimal approximation.

12-S03 Which lengths can ruler and compass actually create? Construct square roots geometrically, then investigate why duplicating a cube requires a different algebraic degree. A constructible real number has degree a power of two over $\mathbb{Q}$; this necessary condition rules out $\sqrt[3]{2}$ [S11, §21.3]. **E:** exact construction questions; **G:** the obstruction proof needs field extensions. “I tried and failed” is not the proof.

Stop/trap: prime-power field size does not mean $\mathbb{Z}/p^n\mathbb{Z}$ is a field. General quintic unsolvability is not “no quintic can be solved.” Full Galois correspondence, inseparability, valuation theory, difference and differential fields remain gated or unscreened. Week 4 cyclic motion is useful preparation but does not itself realize field arithmetic.

FIELD 13 | 4 SURVEYED SEEDS | AD-13; AD-14

13 Commutative algebra

The distinctive objects are rings, ideals, modules, localizations, and dimension: algebra that records the structure of solutions and the failure of familiar cancellation rules. Milne's chapters 1-2 provide the inspected ring/ideal and geometric anchors [S12].

13-S01 What remains after certain monomials are declared zero? On the exponent grid (i,j) for $(x^i y^j)$, mark all multiples of (x^2, xy, y^3) . Ask which monomials survive and how multiplying by (x) or (y) moves or kills them. **E/R:** a monomial quotient with basis $(1, x, y, y^2)$ over a field. Cases and grid movement suffice for discovering the four survivors; vector-space dimension is a later interpretation.

13-S02 Can a nonzero thing square to zero? Compare $(k[t]/(t^2))$ with $(k[t]/(t))$. Track multiplication using $(t^2=0)$ and notice that both equations have the same ordinary zero set, although the quotients differ. **E:** nilpotents and information lost by taking only points. This is a genuine small algebra, not a tiny nonzero real number.

13-S03 What changes when selected numbers become invertible? Allow fractions whose denominators are powers of two. Can $(1/3)$ be represented? Compare with allowing every odd denominator. **E:** localizations of $\mathbb{Z}$. Rational arithmetic permits a complete small argument; prime ideals enter only later.

13-S04 Can infinitely many polynomial constraints be controlled by finitely many generators? First study an upward-closed set of exponent-grid points and its minimal corners. **E:** a monomial-ideal special case. **G:** Hilbert's basis theorem concerns Noetherian coefficient rings and polynomial rings; finitely many drawn corners do not prove it generally.

Stop/trap: quotient by an ideal is not “ignore the inconvenient terms” without checking closure. Gröbner bases, depth, regularity, homological dimensions, and local algebra remain separate research work. Toggle spaces in Week 2 are vector spaces, not automatic commutative-algebra coverage. The monomial grid is a promising genuinely new lower entrance.

FIELD 14 | 4 SURVEYED SEEDS | AD-15; AD-16

14 Algebraic geometry

The territory links equations, geometric objects, maps, singularities, and arithmetic over different fields. Coordinate plotting alone is not enough. Four candidate questions retain distinct geometry:

14-S01 Can every rational point on a circle be found by shooting a rational-slope line from one known point? Use the circle $(x^2+y^2=1)$ and the point $((1,0))$. Solve for the second intersection and account for the exceptional point. **E:** rational parametrization of a conic [S13, §6.5, printed pp. 191-193]. Enter with drawing; solving and proving completeness require fractions and quadratic equations.

14-S02 Where did the second intersection go? Vary a line meeting a parabola: two real intersections can merge or leave the real plane. **E:** line-conic intersection. **G:** Bézout's theorem counts degree-product intersections in the projective plane over an algebraically closed field, with multiplicity and no shared component [S12, Theorem 6.37]. The child experiment should expose missing hypotheses, not claim “all curves always meet that many times.”

14-S03 Do two equations describe the same geometric object? Compare $(y=x^2)$, the parametrization $((t,t^2))$, and $(xy=0)$. Ask whether one parameter can recover every point and whether crossing lines behave like a smooth curve. **E:** specified maps and reducibility; **G:** tangent spaces/singularity schemes need polynomial and linear tools.

14-S04 What does a curve look like over a finite field? Enumerate the solutions of $(y^2=x^3-x)$ over F_5 , then compare with a real plot. **E:** arithmetic points of the same polynomial relation over different fields. This is a point-counting investigation, not yet elliptic-curve group law or cryptography.

Stop/trap: a plotted curve is not a scheme, and a finite grid is not an approximation to all algebraic geometry. Schemes, sheaves, cohomology, moduli, birational geometry, and enumerative geometry remain protected advanced branches. Week 9 rational slopes are relevant preparation, not equivalent content.

FIELD 15 | 4 SURVEYED SEEDS | AD-17; AD-18

15 Linear and multilinear algebra / matrix theory

This field studies combination, dependence, transformation, and canonical structure. It has excellent low entrances, but the familiar toggle model samples only one coefficient field. Axler supplies the linear-map, spectral, and inner-product anchors [S14].

15-S01 Which sets of moves do the same job? For a specified finite toggle board, find both reachable outputs and press patterns that do nothing. **E:** image/kernel over F_2 , with rank-nullity explaining their sizes after a basis is established. Week 2 already owns this mechanism; a new board is only an instance unless the question changes substantially.

15-S02 Which features survive repeated averaging? Apply $((x,y) \mapsto ((3x+y)/4, (x+3y)/4))$. The sum stays fixed while the difference halves. **E:** eigen-directions in a (2×2) map. Children can move two markers and compare midpoints; fractions support exact reasoning. This adds real-scalar dynamics missing from binary toggles.

15-S03 Can a transformation flatten a picture, reverse it, or preserve its area? Compare two planar shears and a projection on a unit-grid parallelogram. **E:** determinant as signed area scaling. Counting squares is entry; signed coordinates and a general proof come later. Do not infer that area preservation implies distance preservation.

15-S04 Which approximate solution is best when exact measurements disagree? Find the nearest point on a line to a marked point, then connect perpendicular error to least squares. **E:** orthogonal projection in a Euclidean inner-product space. Measurement is exploratory; proof requires exact geometry or dot products.

Frontier: multilinear maps, tensors, bilinear forms, canonical forms, matrix inequalities, and spectral algorithms need their own questions. Rank-nullity holds over any field with finite-dimensional domain, but Axler's inspected text restricts its standing scalars to $\mathbb{R}$ or $\mathbb{C}$; the binary version needs its own algebraic proof. No calculus gate is inherent here.

FIELD 16 | 4 SURVEYED SEEDS | AD-19

16 Associative rings and algebras

The distinctive change is that multiplication can encode composition and need not commute. Modules and representations turn algebra into actions on spaces; noncommutativity is not merely an amusing exception. Etingof et al. §§1.2-1.8 and chapters 2, 5 anchor this territory [S15].

16-S01 Does changing the order of two operations change the result? Use matrices ($A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$), ($B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$), and a marked lattice point. **E:** matrix multiplication acting on vectors. Arrow movement allows entry before multiplying arrays; the investigator must specify right-to-left composition.

16-S02 Can two nonzero transformations compose to zero? Project onto the horizontal axis and then project onto the vertical axis. **E:** zero divisors in an endomorphism ring. Contrast with invertible transformations and ask exactly what information disappeared. This is not evidence that real-number multiplication has zero divisors.

16-S03 Can a network of vector spaces be assembled from simpler pieces? Begin with one linear arrow ($V \rightarrow W$). Classify small maps up to independent basis changes by rank, then add a second arrow. **E:** quiver representations. **G:** vector spaces and changes of basis are unavoidable for the mathematical question; dots and arrows without linear data are only a motivational picture.

16-S04 When does an action split into independent parts? Compare the coordinate-axis decomposition of a diagonal operator with a (2×2) Jordan block. **E:** invariant subspaces and failure of complete reducibility. The general semisimplicity question is richer than finding an invariant subset of tokens.

Stop/trap: the set of all functions under composition is a monoid; it becomes an associative algebra only with suitable compatible addition and scalar multiplication. Weeks 3/4 already show noncommuting or cyclic actions but do not cover module decomposition. Division algebras, homological ring theory, noncommutative geometry, PI rings, and quantum groups remain distinct advanced frontier.

FIELD 17 | 3 SURVEYED SEEDS | AD-20

17 Nonassociative rings and algebras

Lie, Jordan, and alternative algebras deserve separate treatment. “Order matters” means noncommutative; nonassociative means parentheses matter. Etingof’s Lie-algebra definition and Baez’s authored octonion survey anchor three different mechanisms [S15, §1.9; S16, §§2-3].

17-S01 Can the failure of two transformations to commute itself be measured algebraically? For real square matrices set $([A,B]=AB-BA)$, and calculate a small commutator table. **E:** a Lie bracket; the Jacobi identity follows by expansion and associativity of the original product. **G:** matrix addition/multiplication and negatives. Physically reversing two finite rotations is motivation, not a direct realization of this infinitesimal bracket.

17-S02 What survives if matrix multiplication is symmetrized? Define $(A \circ B=(AB+BA)/2)$ for real matrices; compare $(A \circ (B \circ C))$ and $((A \circ B) \circ C)$. **E:** a special Jordan algebra and a discoverable failure of associativity. Fractions and matrices are a genuine prerequisite, not something a colorful story removes.

17-S03 Can a seven-point diagram encode multiplication that fails associativity? Use a fixed oriented Fano-plane multiplication convention for octonion units, including signs. Ask for a concrete pair of parenthesizations giving different answers. **E:** multiplication of specified units; reading a signed operation table is sufficient for a finite counterexample. **G:** the normed division algebra, exceptional structures, and their classification need substantial linear algebra.

Stop/trap: never claim any cyclic triangle drawing is “the octonions”; orientation, sign, unit, and anticommutation conventions are essential. A small counterexample establishes failure of associativity, not the division-algebra theorem. The finite table entrance may be mathematically exact yet pedagogically thin; it should compete honestly with the stronger matrix-commutator investigation. No current week substantially covers this field.

FIELD 18 | 4 SURVEYED SEEDS | AD-21; AD-22

18 Category theory and homological algebra

The subject concerns composition, maps between structures, universal constructions, and algebraic information extracted from sequences of maps. Fong-Spivak gives concrete categorical models [S7]; homological algebra must retain its extra linear structure rather than being absorbed into generic arrow drawing.

18-S01 Can two different routes through machines always give the same answer? Supply finite sets, explicit functions, and a square of machines. Test commutativity for every input and repair one map. **E:** commuting diagrams in finite sets. Composition is associative, but equality of the two routes is additional data to investigate.

18-S02 How can two lists be paired using a shared label? Build $\{(a,b):f(a)=g(b)\}$ from two finite labeled collections. Ask why every other consistently paired collection has exactly one map to this collection preserving its two projections. **E:** pullback in sets [S7, §3.5.3]. Objects supplies the construction; the uniqueness property needs Structural reasoning.

18-S03 What is the best translation between “resources required” and “tasks possible”? The benches/children adjunction in field 06 is an exact preorder example. The shared mechanism should be one family with two research links, not two counted activities.

18-S04 Which closed edge-patterns are boundaries of faces? Work over F_2 with vertices, edges, and triangular faces. The boundary of a face-boundary is zero; a triangle loop is a cycle whether or not its face is filled, but it is a boundary only in the filled version. **E/R:** a small chain complex and its first homology. Source support for the general homological setting is the Stacks Project's complexes and cohomology section [S17]; the finite construction needs its own checked matrices.

Stop/trap: a loop in a picture need not represent a nonzero homology class. A commuting diagram is not automatically a universal one. Derived categories, higher categories, operads, toposes, and spectral sequences remain hard gates. Week 2's incidence map is an exact preparatory object; adding faces and quotienting cycles by boundaries creates the new mathematical question.

FIELD 19 | 4 SURVEYED SEEDS | AD-23

19 Algebraic K-theory

K-theory builds invariants from algebraic or geometric objects and how they combine; stable equivalence and group completion are genuine initial mechanisms, but not the whole field. Weibel II §§1-2 was inspected directly [S18].

19-S01 Can we consistently subtract collections that only know how to combine? Represent a formal difference by a pair (a,b) ; for ordinary nonnegative counts, identify (a,b) with (c,d) when $(a+d=b+c)$. **E:** group completion of $\mathbb{N}$ to $\mathbb{Z}$. Counters and cancellation permit early entry. The interesting question is well-definedness of addition and equivalence, not routine negative-number arithmetic.

19-S02 What if cancellation fails? Try the commutative monoid $\{0,e\}$ with $(e+e=e)$. In any group receiving a monoid homomorphism, the image of (e) must be zero. **E:** group completion can lose information. This counterexample guards against pretending the simple pair rule describes all monoids without adjustment; general equality allows adding a common extra element.

19-S03 Can two kinds of symmetry-orbit be counted independently? For a two-element group, classify finite actions into fixed points and swapped pairs. Addition is disjoint union. **E:** the additive monoid underlying its Burnside ring, whose completion is $\mathbb{Z}^2$; investigate products to see why total number of points alone forgets structure [S18, Example 1.5].

19-S04 Why is $(K_0(k) \cong \mathbb{Z})$ for a field? Finite-dimensional vector spaces combine by direct sum and are classified by dimension. **G/E:** this is actual (K_0) , once vector spaces/projective modules and the definition are available. The counter story in question 1 is its underlying completion mechanism, not a claim that all counter play teaches algebraic K-theory.

Stop/trap: higher (K) -groups, projective modules over general rings, vector bundles, localization theorems, and links to arithmetic require protected advanced work. This field should be marked “small exact construction found; major prerequisite gate remains,” not rejected wholesale or declared elementary.

FIELD 20 | 4 SURVEYED SEEDS | AD-24

20 Group theory and generalizations

Symmetry, generation, actions, quotients, representations, and geometric group theory provide different questions. The following extend beyond merely identifying repeated cycles; Judson §§14.1 and 14.3 anchor action and orbit counting [S11].

20-S01 How many moves does it take to restore a labeled triangle? Generate its six symmetries using a rotation and one reflection; compare words that act identically. **E/R:** (D_3) , generators/relations, and a finite Cayley graph. Objects permits manipulation; Cases is needed to establish completeness and shortest words. Contrast the two orders of rotation and reflection.

20-S02 How many genuinely different necklaces are there? For length-four binary necklaces, decide explicitly whether only rotations or also reflections identify arrangements. Count orbits directly, then by fixed configurations. **E:** finite group action; Burnside averages fixed-point counts. Dividing by the number of symmetries fails when stabilizers vary.

20-S03 What information can be forgotten while multiplication still makes sense? Send a permutation to its parity, or a square symmetry to its action on the two diagonals. **E:** a homomorphism and its kernel. The question is which distinctions can be erased consistently, not merely grouping objects by resemblance.

20-S04 What changes when moves are not reversible? Add a “send every token to one place” operation. **E:** transformation semigroups/monoids, which are included in this MSC field. Compare eventual cycles and transient behavior with the pure permutation case. This changes the mathematical question from Week 3, where every state lies on a cycle.

Access: the first two can begin with spatial operations and systematic cases; parity needs transpositions and a proof that the sign is well-defined. **Stop/trap:** symmetry group, configuration space, and set of legal moves are different objects. Representations require linear structure. Infinite groups, growth, word problems, group cohomology, profinite groups, and classification of finite simple groups remain separate frontier. Weeks 3/4 are direct overlap; the most promising new choices are stabilizers, quotient maps, and irreversible transformations.

Priorities and non-duplication decisions

The next bridge-design batch should prioritize antichains; tree codes; closure systems; valid versus invalid quotienting; triangular-square numbers; four-symbol field arithmetic; monomial staircases; rational conics; averaging eigen-directions; and face boundaries. These offer different decisions and explanations from current Weeks 1-10.

A protected advanced batch should tackle polynomial/field obstructions, quiver decompositions, matrix commutators, and group completion with failure of cancellation. It is acceptable for those to stop at explicit algebra or linear-algebra gates. Historical comparison and proof/model questions should be tagged as cross-cutting lenses; they must not pad the family count.

No statement here justifies “all of algebra surveyed exhaustively.” Every listed top-level field has been substantively screened; its internal frontier remains large. Recommended ledger dispositions are **00/01: facet with substantive candidate questions, 03/05/06/08/11-18/20: candidate mechanisms, review pending, 19: exact preliminary mechanisms plus identified advanced gates**. None is “classroom validated.”

Inspected source register

URLs below were opened through the web tool on 25 September 2026; specific theorem/section text was inspected where cited. These sources support mathematical anchors, not the proposed capability profiles. No downloaded source files or redistributed book text were created by this survey.

- **S0.** AMS/zbMATH, [MSC2020 database](#) and [research-taxonomy help](#). Supports scope and taxonomy purpose; the coordinator owns the complete pinned import.
- **S1.** Euclid, *Elements*, David E. Joyce's edition, [VII.2](#): common measure by repeated subtraction. Translation/commentary, not a claim to inspect an ancient manuscript.
- **S2.** Euclid/Joyce, [IX.20](#): prime outside any assigned finite list; inspect the prime/composite split.
- **S3.** Euler Archive, [E53: Solutio problematis ad geometriam situs pertinentis](#), archival metadata and content summary. Original-paper text was not inspected; any comparison of Euler's exact presentation remains an additional task.
- **S4.** Open Logic Project, [Sets](#), [Logic](#), [Computation](#), chapter 4 diagonalization (printed pp. 58-60), §12.9/Theorem 12.23 compactness, chapter 15 undecidability. Inspected corresponding PDF text; no source claim about young learners.
- **S5.** Richard P. Stanley, [Topics in Algebraic Combinatorics](#), [1 February 2013 author draft](#): Theorem 1.1 walks, chapter 4 Sperner property, Theorem 9.8 Matrix-Tree, chapter 9 appendix on tree encodings. This PDF's printed pagination differs from its file page count and published editions.
- **S6.** Stanley Burris and H. P. Sankappanavar, [A Course in Universal Algebra](#), Millennium edition: I §§1-5; II §§5-11, especially Theorem 11.9 (Birkhoff); III §§1-4. The authors' [distribution page](#) describes editions and rights.
- **S7.** Brendan Fong and David I. Spivak, [Seven Sketches in Compositionality](#), Definition 1.95 and Example 1.97 (Galois connections); §3.5.3, pullbacks of sets, especially colored-pair example; chapter 3 categories/functions. The benches interpretation is this survey's proposed finite-count specialization.
- **S8.** Keith Conrad, [The Chinese Remainder Theorem](#), Theorem 1.1 and §2 proof.
- **S9.** Keith Conrad, [Pell's Equation, I](#), §3/Theorem 3.1 triangular-square numbers; §4 multiplication of quadratic units.
- **S10.** Keith Conrad, [Finite Fields](#), Theorem 1.1, examples and warning on p. 1; executive summary. Four-element model here is a proposed specialization.
- **S11.** Thomas W. Judson, [Abstract Algebra: Theory and Applications](#), §21.3 Geometric Constructions, §14.1 Groups Acting on Sets, §14.3 Burnside's Counting Theorem. Also inspected chapter-level contents for fields, finite fields, and lattices.
- **S12.** J. S. Milne, [Algebraic Geometry](#), version 6.10, chapter 1 rings/ideals/localizations; chapter 2 Hilbert basis theorem and Theorem 2.16 Nullstellensatz; Theorem 6.37 Bézout. Assumptions about the field must be carried with theorem statements. The specific monomial quotient and finite-field enumeration are proposed examples, not attributed exercises.
- **S13.** Ravi Vakil, [The Rising Sea](#), January 29, 2015 draft, §6.5, printed pp. 191-193, rational parametrization of a circle. Inspected historical draft, explicitly not the current published edition.

- **S14.** Sheldon Axler, *Linear Algebra Done Right*, fourth edition author PDF, §3A linear maps/composition, §3B/Theorem 3.21 rank-nullity, §6C/Definition 6.55 orthogonal projection, §7B/Theorems 7.29 and 7.31 spectral theorems. Source standing scalars are real/complex; do not cite it as if its text states the finite-field version.
- **S15.** Pavel Etingof et al., *Introduction to Representation Theory*, January 10, 2011 notes, §§1.2-1.9 algebras/representations/quivers/Lie algebras, chapter 2 decompositions, chapter 5 quivers. Definition 1.39 and Example 1.40 explicitly inspect the Lie bracket and Jacobi identity. These notes use numbering different from the later book.
- **S16.** John C. Baez, *The Octonions*, author PDF, §2.1 Fano-plane rules (printed pp. 6-8) and §3 Jordan product/discussion (printed pp. 19-20); also inspected the [associator discussion](#). PDF text was inspected. A final signed-table activity must visually inspect the multiplication diagram before selecting an exact instance.
- **S17.** Stacks Project, §12.13 “Complexes”, definition of homology following Lemma 12.13.3 and the cochain counterpart; also inspected Lemma 12.13.12. This is not a source for a child activity; a future design must explicitly verify its finite boundary maps and quotient.
- **S18.** Charles Weibel, *The K-book*, chapter II, §1 group completion, Proposition 1.1, Example 1.5 Burnside ring; §2 definition and first computations of (K_0) . [Author contents page](#) distinguishes chapter-PDF pagination from the published volume.

PART II

Geometry, topology, and analysis

Prepared 25 September 2026. This is a **research-screening document**, not a collection of reviewed lessons. It covers the 27 assigned MSC2020 top-level fields individually. Three or four anchors per field establish useful starting territory; they do not exhaust the field or confer dispositions on its descendant codes. Proposed entrances are design inferences, with no classroom evidence claimed. The advisory `math_circle_atlas_plan.md`, project README, current fall plan, and Weeks 2-10 mathematical redesign were consulted.

The survey deliberately includes mathematics whose first satisfactory entrance requires calculus, linear algebra, or university analysis. Lowering every topic would damage the map. Conversely, continuous motion, comparison, and geometric arguments sometimes provide genuine access before numerical calculation does.

Physical measurement suggests a conjecture; an idealized model and argument establish a theorem. An apparent crossing in a graph, a measured right angle, or one successful soap film does not resolve the corresponding universal claim.

Access descriptions below are capability gates, not age placement. “Entry” means meaningful work on the stated small world; it does not imply readiness to prove its adult generalization. Unless otherwise specified, materials are paper, string, movable markers, transparent overlays, or a simple graphing interface. Sources are authored mathematical texts, institutional lecture notes, or the NIST reference; locators identify what was inspected. Calculations explicitly called “small exact examples” are independently specified here, rather than borrowed lesson designs.

FIELD 22 | 4 SURVEYED SEEDS | GA-01

22 Topological groups, Lie groups

What distinguishes the field. Symmetry has continuous parameters, local geometry, and infinitesimal generators. Classifying finite permutations alone misses this territory.

Question seeds.

22-S01 Plane rotations about one fixed center obey $R(a)R(b)=R(a+b)$ for all real angles; the parameter is taken modulo a full turn.

22-S02 Rotations about different axes in three dimensions generally do not commute: quarter-turns about the x- and y-axes already distinguish the two orders.

22-S03 For real skew-symmetric matrices A , $\exp(tA)$ is orthogonal with determinant one; differentiation recovers the generator.

22-S04 Unit quaternions act on imaginary quaternions by conjugation, giving the two-to-one map onto $SO(3)$, with kernel $\{1, -1\}$. Woit, *Quantum Theory, Groups and Representations*, chapters 4-6.

Entrances and stops. Rotate an arrow continuously, combine chosen turns, and ask whether order matters. Then manipulate a marked box using two specified axes. This is an exact action of rotations, requiring orientation tracking but no reading beyond instructions. Matrix exponentials stop at linear algebra plus convergent power series; the covering-space claim stops at topology. A belt demonstration is motivation until its allowed deformations are modeled precisely.

Trap/frontier. A finite set of quarter-turns does not exhibit Lie algebra or local topology. Retain representation theory, noncompact groups, and harmonic analysis on groups as separate later searches. Week 3 permutations are a useful contrast, not coverage of this field.

FIELD 26 | 4 SURVEYED SEEDS | GA-02; GA-28

26 Real functions

Distinguishing mathematics. Completeness, continuity, differentiability, and controlled limits turn pictures of change into existence and impossibility arguments.

Question seeds.

26-S01 A continuous real function on $[a,b]$ assumes every value between its endpoint values.

26-S02 Such a function attains its maximum and minimum.

26-S03 If continuous on $[a,b]$ and differentiable on (a,b) , it has a point where instantaneous slope equals the secant slope.

26-S04 Continuous functions can converge pointwise to a discontinuous function: x^n on $[0,1]$; uniform convergence would preserve continuity. Lebl, *Basic Analysis I*, §§3.3, 4.2, 6.1-6.2.

Entrances and stops. Two markers exchange which is higher while moving without jumps; explain why equality is unavoidable. Give a continuous periodic fence profile and compare opposite points: after half a turn their height difference changes sign. This is an exact continuous special case, not an exercise about six isolated posts. Numerical bisection adds fractions and gives an interval-width guarantee. Moving averages and speed stories introduce slopes, but the mean-value theorem stops at a derivative concept.

Trap/frontier. “No pencil lifting” is an unreliable definition of continuity, and a steep continuous ramp is not a jump. Keep nowhere-differentiable functions, bounded variation, generalized derivatives, and multivariable inequalities on the frontier.

FIELD 28 | 4 SURVEYED SEEDS | GA-03; GA-29

28 Measure and integration

Distinguishing mathematics. Size is assigned consistently to sets, and integrals survive suitable limiting operations; counting points and measuring length can disagree spectacularly.

Question seeds.

28-S01 A countable subset of the real line has Lebesgue measure zero: cover its n th point by an interval of length at most $\varepsilon/2^n$.

28-S02 The middle-thirds Cantor set is uncountable but has length zero; its n th cover has length $(2/3)^n$.

28-S03 For measurable $0 \leq f_n \uparrow f$, integrals increase to the integral of f , allowing infinity.

28-S04 For measurable f_n with a fixed integrable bound $|f_n| \leq g$, almost-everywhere convergence permits exchanging integral and limit. *Axler, Measure, Integration & Real Analysis*, §§2A, 2D, 3A-3B.

Entrances and stops. Cover finitely many named points with arbitrarily short paper strips, then design a rule covering a listed infinity. Repeated cutouts introduce nested sets; fractions and infinite series support the zero-length argument, while a separate binary-sequence argument establishes uncountability. This is a strong secondary-level entrance; younger finite constructions are preparation.

Trap/frontier. Finite dust still has positive paper area; small pixels are not points. A zero-area object need not be empty. No finite cutting activity proves nonmeasurability. Product measures, Hausdorff measure/dimension, differentiation of measures, and nonmeasurable sets need separate searches.

FIELD 30 | 4 SURVEYED SEEDS | GA-04

30 Functions of a complex variable

Distinguishing mathematics. Complex differentiability imposes rigidity: local behavior can determine a whole function, while contour geometry controls zeros and integrals.

Question seeds.

30-S01 Multiplication by nonzero $re^{i\theta}$ scales all lengths by r and rotates all angles by θ .

30-S02 A nonconstant holomorphic function on a connected open set has no interior local maximum of its modulus.

30-S03 For a function holomorphic on a neighborhood of a closed Jordan region, with no zeros on its positively oriented boundary, the change in argument counts enclosed zeros with multiplicity.

30-S04 $z \mapsto z^2$ doubles argument and squares modulus, so a full target circuit exchanges the two square-root branches. Lebl, *Guide to Cultivating Complex Analysis*, §§1.2, 3.3, 5.1 and chapter 10.

Entrances and stops. Rotating and stretching arrows gives an exact model of multiplication. Track a two-valued square root around a circle with two colored moving markers; algebra and angle doubling suffice for that special case. Contour integrals and the general argument principle stop at complex differentiation/integration.

Trap/frontier. Pretty complex-plane coloring does not establish holomorphicity, and angle preservation excludes points where the derivative vanishes. Riemann surfaces, analytic continuation, conformal mapping, and value distribution remain distinct territory.

FIELD 31 | 4 SURVEYED SEEDS | GA-05; GA-35

31 Potential theory

Distinguishing mathematics. Interior values are governed by boundary data and averaging, connecting equilibrium, harmonicity, and energy.

Question seeds.

31-S01 A C^2 harmonic function has its center value equal to its average over every contained sphere or ball.

31-S02 A continuous function with this local mean-value property is smooth and harmonic.

31-S03 On a bounded connected domain, a harmonic function continuous on the closure cannot exceed its boundary maximum; an interior maximum forces constancy. These hypotheses and the reverse direction are inspected in [Hunter](#), [Notes on PDE](#), §§2.1 and 2.3, [Theorems 2.1-2.2](#).

31-S04 A small exact discrete counterpart: on a finite connected graph with nonempty fixed boundary, interior vertices equal to their neighbor averages have at most one solution, because a positive difference maximum propagates to the boundary.

Entrances and stops. Invent number landscapes satisfying the averaging rule; ask whether an isolated peak is possible. Four-neighbor graph models require addition and division. A continuous rubber-sheet or temperature picture is an analogy unless the governing PDE and boundary assumptions are justified. Real harmonic examples x , x^2-y^2 , and $\log r$ away from zero require elementary differentiation.

Trap/frontier. Graph harmonicity is exact mathematics but does not prove convergence to a continuum solution. Capacity, Green functions, fine topology, and nonlinear potential theory retain hard analysis gates.

FIELD 32 | 3 SURVEYED SEEDS | GA-06

32 Several complex variables and analytic spaces

Distinguishing mathematics. Multiple complex coordinates produce phenomena absent in one variable; they are not just two simultaneous complex drawings.

Question seeds.

32-S01 For $n \geq 2$, if U is a domain in $\mathbb{C}^n$, K is compact in U , and $U \setminus K$ is connected, every holomorphic function on $U \setminus K$ extends holomorphically to U (Hartogs).

32-S02 The unit ball and unit bidisc in $\mathbb{C}^2$ are not biholomorphic.

32-S03 A nonempty zero set of a holomorphic scalar function on a domain in $\mathbb{C}^n$, $n \geq 2$, cannot be compact. [Lebl](#), *Tasty Bits of Several Complex Variables*, Theorem 4.3.1, Corollary 4.3.2, §1.4/Theorem 1.4.4.

Entrances and stops. Compare $1/z$ on a punctured plane with the impossibility of an isolated holomorphic pole in a punctured ball in $\mathbb{C}^2$. A meaningful investigation can start with polynomial examples such as the zero set of zw , then ask which one-variable intuitions survive. Polynomial algebra permits examples, but these distinguishing extension results need one-variable complex analysis and multivariable reasoning. **No elementary bridge established.**

Trap/frontier. Filling a physical hole is not Hartogs extension. Slicing a four-real-dimensional space into pictures may hide compatibility conditions. Pseudoconvexity, the $\bar{\partial}$ problem, sheaves, singularities, and analytic spaces deserve their own advanced dossiers.

FIELD 33 | 3 SURVEYED SEEDS | GA-07

33 Special functions

Distinguishing mathematics. Recurrences, differential equations, integral representations, and symmetries organize functions that recur throughout mathematical physics and geometry.

Question seeds.

33-S01 Chebyshev polynomials satisfy $T_n(\cos \theta) = \cos(n\theta)$, with $T_0=1$, $T_1=x$, and $T_{n+1}=2xT_n - T_{n-1}$.

33-S02 The gamma function extends factorial values: for positive real x , $\Gamma(x+1)=x\Gamma(x)$ and $\Gamma(1)=1$.

33-S03 Bessel's equation $x^2y''+xy'+(x^2-v^2)y=0$ supplies radial modes for circular geometry, rather than ordinary sine waves. Exact references: [NIST DLMF 18.5.1](#) and [§18.9, 5.5.1, 10.2.1](#).

Entrances and stops. Use angle-doubling circles to discover $T_2=2x^2-1$, then obtain higher polynomials in two ways. This requires coordinates/algebra; a gear drawing alone gives only motivation. Compare candidate “factorial between integers” rules to see why a recurrence does not determine a unique interpolation. Gamma's integral and Bessel's equation stop at calculus; drumhead mode proofs need PDE.

Trap/frontier. Naming a curve after a special function is not investigating its mechanism. Orthogonality, asymptotics, hypergeometric transformations, and elliptic functions remain separate families.

FIELD 34 | 4 SURVEYED SEEDS | GA-08; GA-34

34 Ordinary differential equations

Distinguishing mathematics. A local rate law can determine an entire trajectory, but existence, uniqueness, and long-time survival are different questions.

Question seeds.

34-S01 A continuous vector field locally Lipschitz in the dependent variable gives a unique local initial-value solution. Teschl, *Ordinary Differential Equations and Dynamical Systems*, §2.2, Theorem 2.2. Small exact examples:

34-S02 $y'=y^2$, $y(0)=1$ has $y=1/(1-t)$ only up to its finite blow-up time.

34-S03 $y'=2\sqrt{y}$, $y(0)=0$, restricted to nonnegative y and forward time, permits waiting until any $c\geq 0$ then taking $y=(t-c)^2$.

34-S04 $y'=y(1-y)$ has equilibria 0 and 1, with upward motion between them and downward motion above 1.

Entrances and stops. Place direction arrows along a number line and predict trajectories before solving formulas. Rate comparisons and signed numbers support qualitative reasoning; checking formulas requires derivatives. The waiting solution provides a surprising calculus-level investigation with an explicit failure of the uniqueness hypothesis.

Trap/frontier. Connecting Euler-method dots gives an approximation, not the solution; small steps can still distort stability. Systems, bifurcation, Sturm-Liouville theory, boundary-value problems, and stiff equations require further work.

FIELD 35 | 4 SURVEYED SEEDS | GA-09

35 Partial differential equations

Distinguishing mathematics. Change depends on several independent variables; different equations propagate information in fundamentally different ways.

Question seeds.

35-S01 For C^2 profiles F and G , $u(x,t)=F(x-ct)+G(x+ct)$ solves the one-dimensional wave equation $u_{tt}=c^2u_{xx}$.

35-S02 On $[0,\pi]$, $u=e^{-t}\sin x$ solves $u_t=u_{xx}$ with zero endpoint values, retaining its shape while decaying.

35-S03 A small exact Burgers example: $u(x,t)=-x/(1-t)$ solves $u_t+uu_x=0$ for $t<1$ with initial profile $u(x,0)=-x$; characteristics $x(t)=\xi(1-t)$ collide at $t=1$.

35-S04 The harmonic maximum principle provides uniqueness for continuous Dirichlet data whenever a sufficiently regular solution exists. Hunter, *Notes on PDE*, §§2.3, 5.1, and 7.1; the displayed Burgers example is checked directly by differentiation.

Entrances and stops. Slide two drawn wave profiles past one another and add heights; this preserves exact solutions when the profiles are smooth. Compare the wave and heat examples using calculus. Traffic trajectories introduce characteristic crossing with coordinate geometry, but choosing physically meaningful weak solutions needs additional analysis.

Trap/frontier. “Everything spreads out” conflates waves, diffusion, and transport. A classroom token-sharing rule is a discrete model, not automatically a heat equation. Nonlinear regularity, shocks, distributions, and multidimensional PDE remain large open design territory.

FIELD 37 | 4 SURVEYED SEEDS | GA-10

37 Dynamical systems and ergodic theory

Distinguishing mathematics. Repeated evolution produces orbits, invariant sets, stability, recurrence, and statistical behavior; unpredictability can coexist with exact rules.

Question seeds.

37-S01 Circle rotation by a rational fraction of a turn is periodic; an irrational rotation has dense orbits.

37-S02 For $f_r(x)=rx(1-x)$, the fixed point $1-1/r$ is locally attracting for $1<r<3$, since its derivative is $2-r$.

37-S03 The tent map $T(x)=2x$ below $1/2$ and $2-2x$ above it stretches intervals and folds them; finite itineraries can be built by inverse branches.

37-S04 A probability-preserving transformation on a finite measure space returns almost every point of a measurable set to it infinitely often. For iteration, symbolic dynamics, and attractors, see [Teschl, chapter 11](#), especially §§11.3-11.5; recurrence is an advanced frontier anchor requiring a dedicated proof source before lesson production.

Entrances and stops. Iteration games and cobweb plots require following a rule; rational arithmetic supports exact orbit certificates. Irrational density needs an infinite argument. Ergodic conclusions stop at measure theory.

Trap/frontier. A long nonrepeating decimal simulation proves neither irrationality nor chaos. Do not label all nonlinear iteration chaotic. Week 4 modular orbits and Week 9 billiards are prior overlaps; invariant measures and continuous flows provide genuinely new questions.

FIELD 39 | 4 SURVEYED SEEDS | GA-11

39 Difference and functional equations

Distinguishing mathematics. Global rules constrain sequences or entire functions, often leaving freedom that sample values conceal.

Question seeds. Small exact examples:

39-S01 Every solution on nonnegative integers of $a_{n+2} - 2a_{n+1} + a_n = 0$ is $a_n = A + Bn$.

39-S02 For $a_{n+1} = ra_n + b$, subtracting a fixed point reduces the problem to a geometric sequence when $r \neq 1$.

39-S03 Additivity $f(x+y) = f(x) + f(y)$ on $\mathbb{Q}$ forces $f(q) = qf(1)$. On $\mathbb{R}$ this conclusion follows with continuity at one point, but fails without suitable regularity.

39-S04 A continuous positive function satisfying $f(x+y) = f(x)f(y)$ on $\mathbb{R}$ is exponential. The regularity problem is discussed in Reem, *Remarks on the Cauchy functional equation and variations of it*, introduction and §2.

Entrances and stops. Let children invent machines obeying “combine inputs, add outputs,” then predict untested values and explain necessity. Whole-number versions need only addition; rational extension needs division. Distinguishing rational from real conclusions needs continuity and infinite reasoning.

Trap/frontier. Agreement on a finite table does not establish an equation for every input. Uniqueness requires a domain and regularity assumptions. Nonlinear recurrences, delay equations, and stability of approximate functional equations remain separate searches.

FIELD 40 | 4 SURVEYED SEEDS | GA-12

40 Sequences, series, summability

Distinguishing mathematics. Infinite addition has rules that finite arithmetic cannot safely supply; summation procedures differ from ordinary convergence.

Question seeds. Small exact examples:

40-S01 $1+1/2+1/4+\dots=2$, because the partial-sum remainder is an explicit power of $1/2$.

40-S02 The harmonic series diverges by grouping blocks whose sum is at least $1/2$.

40-S03 The ordinary partial sums of $1-1+1-1+\dots$ alternate, but their arithmetic means tend to $1/2$.

40-S04 Every convergent sequence has arithmetic means converging to the same limit; the converse fails.

General series rules and convergence hypotheses: Lebl, *Basic Analysis I*, §§2.5-2.6. Fourier summability gives a distinct application in Stroock, *Topics in Fourier Analysis*, §5.

Entrances and stops. Build area dissections, accumulate strips, and compete to exceed a target using harmonic terms. Fractions and grouping support exact proofs. Ask what an averaging rule measures before calling its output a “sum.”

Trap/frontier. A bounded sequence need not converge; terms tending to zero are necessary but insufficient for convergence of a series. Never use regularized sums as ordinary sums. Conditional rearrangements, Tauberian conditions, and summability matrices need dedicated later dossiers.

FIELD 41 | 4 SURVEYED SEEDS | GA-13

41 Approximation and expansions

Distinguishing mathematics. An approximation is judged by a specified error, domain, and class of allowable approximants; matching many sample points is only one possible objective.

Question seeds.

41-S01 For continuous real f on a compact nondegenerate interval, there is a unique best uniform approximating polynomial of degree at most n . Alternating maximal error at $n+2$ ordered points certifies optimality. [NIST DLMF §3.11\(i\)](#). Small exact examples:

41-S02 The best constant approximation to x^2 on $[-1,1]$ is $1/2$, error $1/2$: values at 0 and 1 force that lower bound.

41-S03 The best uniform affine approximation to x^2 is also $1/2$, by symmetry and the same bound.

41-S04 Linear interpolation of a C^2 function on an interval of length h has error at most $h^2 \sup|f''|/8$.

Entrances and stops. Move a horizontal strip until a drawn curve fits in the narrowest strip; then justify endpoints forcing the answer. This needs comparison before numbers. Secondary learners compare worst error with squared error, then build counterexamples where more interpolation points make behavior worse.

Trap/frontier. “Looks close” is not an error certificate. Numerical point samples can miss peaks between them. General interpolation bounds require calculus; approximation in function spaces, rational approximation, splines, and wavelets deserve distinct attention.

FIELD 42 | 4 SURVEYED SEEDS | GA-14

42 Harmonic analysis on Euclidean spaces

Distinguishing mathematics. Decomposing functions into oscillations reveals scale, cancellation, regularity, and uncertainty.

Question seeds.

42-S01 Integer-frequency complex exponentials are orthogonal on a full unit period.

42-S02 For integrable functions on the circle, Fourier coefficients of their circular convolution multiply.

42-S03 Fejér means of a continuous periodic function converge uniformly to that function, even when ordinary partial sums are not the appropriate convergence method.

42-S04 Two continuous oscillations whose frequencies differ by an integer multiple of the sampling rate give the same equally spaced samples. Orthogonality, convolution, and Fejér's positive averaging kernel: [Stroock, *Topics in Fourier Analysis*, §§1-5, especially formula \(5.1\).](#)

Entrances and stops. Combine rotating arrows and draw their height; change amplitude and phase to match a target. Sample a turning wheel with a fixed strobe and exhibit aliasing exactly through angle arithmetic. Trigonometry supports finite sums; infinite reconstruction and convergence require integration and analysis.

Trap/frontier. A finite DFT exactly recovers a finite vector, not every continuous signal compatible with it. Sound demonstrations motivate but do not prove Fourier completeness. Singular integrals, maximal functions, restriction, and time-frequency analysis must not disappear behind a music activity.

FIELD 43 | 4 SURVEYED SEEDS | GA-15

43 Abstract harmonic analysis

Distinguishing mathematics. Frequencies are characters or representations of an underlying group; the group's structure determines the transform.

Question seeds.

43-S01 For a finite abelian group, its complex characters form an orthogonal basis of functions on the group, with normalized counting inner product.

43-S02 Translation becomes multiplication by character values under Fourier transformation.

43-S03 On $(\mathbb{Z}/2\mathbb{Z})^n$, characters are sign patterns $(-1)^{a \cdot x}$, producing the Walsh transform.

43-S04 For the circle group, continuous characters are $z \mapsto z^n$, indexed by integers, so the dual is discrete rather than another circle. Dandavati, *The Fourier Transform on Finite Groups: Theory and Computation*, §3, Theorems 3.5-3.8 and Remark 3.7.

Entrances and stops. Four signed cards can encode a four-entry brightness pattern as average, left-right, top-bottom, and checkerboard components. Children choose a pattern to hide and reconstruct. Signed addition and halves give an exact finite case. Characters require groups and complex numbers; Haar integration and nonabelian representation theory are hard stops.

Trap/frontier. Fourier-on-four-cards overlaps linear algebra and Week 2 toggles but contributes a new question only if translation, decomposition, or reconstruction is investigated. It does not review locally compact groups, amenability, or noncommutative harmonic analysis.

FIELD 44 | 4 SURVEYED SEEDS | GA-16; GA-33

44 Integral transforms, operational calculus

Distinguishing mathematics. Transforming a function can turn evolution or convolution into algebra, but inversion depends on information and analytic conditions.

Question seeds.

44-S01 For piecewise continuous functions of exponential order, one-sided Laplace transforms exist in a suitable right half-plane.

44-S02 With appropriate differentiability and exponential bounds, $L(f') = sL(f) - f(0)$.

44-S03 Causal convolution transforms into multiplication where the integrals converge absolutely. These rules and regions of convergence are explicit in [Boyd, EE102 Lecture 3, slides 3-2 through 3-7 and 3-27 through 3-29](#). Small exact example:

44-S04 Convolving two unit interval indicator functions produces a triangular tent supported on $[0, 2]$.

Entrances and stops. Slide two paper intervals past one another and record overlap length against displacement. This is a literal convolution integral and needs only length comparison for entry; derive the piecewise formula with coordinates. An impulse-response “echo machine” with finite sums is a discrete analogue. Laplace solving of ODEs stops at integration plus algebra.

Trap/frontier. Transform tables hide domains; an algebraic formula without its region of convergence can be ambiguous. A finite collection of shadows does not automatically determine an object. Radon/Mellin transforms, inverse problems, and distributional operational calculus need further scouting.

FIELD 45 | 4 SURVEYED SEEDS | GA-17

45 Integral equations

Distinguishing mathematics. The unknown function appears inside an averaging or accumulation operator; some kernels reduce an infinite-dimensional question to finitely many unknown moments.

Question seeds. Small exact examples on $[0, 1]$:

45-S01 $u(x) = 1 + \lambda \int u(t) dt$ has constant solution $1/(1-\lambda)$ for $\lambda \neq 1$ and no solution for $\lambda = 1$.

45-S02 $u(x) = \lambda \int u(t) dt$ has only zero for $\lambda \neq 1$ and every constant for $\lambda = 1$.

45-S03 $u(x) = 1 + \int_0^x u(t) dt$ gives $u = e^x$; differentiating is justified once u is continuous.

45-S04 For compact K on a Hilbert space, $I + K$ is invertible exactly when its nullspace is zero; otherwise solvability requires orthogonality to the adjoint nullspace. O'Neil, *Integral Equations and Fast Algorithms*, Fredholm Alternative, Theorem 3, printed p.54.

Entrances and stops. Use “your value equals a gift plus a fraction of the group's mean” puzzles, then replace finitely many participants by a continuous interval. The first three examples remain genuine calculus-level tasks; the finite group version is explicitly a matrix model.

Trap/frontier. A Riemann sum creates an approximate equation, potentially with different stability. No elementary realization of general Fredholm theory is claimed. Singular kernels, ill-posed first-kind equations, and boundary integral methods remain open searches.

FIELD 46 | 4 SURVEYED SEEDS | GA-18

46 Functional analysis

Distinguishing mathematics. Functions become points in spaces; geometry and completeness govern approximation, continuity, and existence.

Question seeds.

46-S01 A nonempty closed convex subset of a Hilbert space has a unique nearest point to each point of the ambient space.

46-S02 Projection onto a closed linear subspace is characterized by orthogonality of the residual.

46-S03 A contraction of a nonempty complete metric space has a unique fixed point; successive approximations converge geometrically.

46-S04 A bounded linear functional on a Hilbert space is inner product with one unique vector. [Axler, *Measure, Integration & Real Analysis*](#), chapter 8, especially closed subspaces and Riesz representation; [Lebl, *Basic Analysis I*](#), §7.6.

Entrances and stops. Find a closest point on a line using string and a right-angle argument; then compare “distance between graphs” measured by maximum discrepancy or integrated squared discrepancy. The first is a finite-dimensional exact special case. Infinite-dimensional geometry needs norms, completeness, and often integration.

Trap/frontier. Changing the norm can change the best answer and destroy uniqueness. Closed bounded sets are not generally compact in infinite dimensions. Banach-space duality, weak topology, distributions, and locally convex spaces need separate mechanisms; one closest-point activity cannot represent them all.

FIELD 47 | 4 SURVEYED SEEDS | GA-19

47 Operator theory

Distinguishing mathematics. Study transformations of whole function spaces, including spectra, invertibility, compactness, and long-term repeated action.

Question seeds.

47-S01 Compact normal operators on complex Hilbert spaces have orthonormal eigenvector bases. Axler, *Measure, Integration & Real Analysis*, Theorem 10.107. Small exact contrasts:

47-S02 The right shift on ℓ^2 , $(a_0, a_1, \dots) \mapsto (0, a_0, a_1, \dots)$, preserves norm but is not onto.

47-S03 Differentiation on polynomials is linear but unbounded in the sup norm on $[0, 1]$, since $\|x^n\|_\infty = 1$ and $\|(x^n)'\|_\infty = n$.

47-S04 If bounded T has operator norm below one, the geometric operator series gives $(I - T)^{-1} = \sum T^n$.

Entrances and stops. Explore a finite “blur machine,” ask which brightness patterns keep their shape, then predict repeated blurring from eigenvectors. Signed arithmetic starts a matrix special case. The shift and unbounded-derivative contrasts need infinite sequences/calculus and should retain those gates.

Trap/frontier. Every finite matrix has finite-dimensional spectral behavior; it cannot demonstrate continuous spectrum. Not every operator is diagonalizable. Operator algebras, non-self-adjoint spectra, semigroups, and unbounded operator domains remain explicitly unbridged.

FIELD 49 | 4 SURVEYED SEEDS | GA-20

49 Calculus of variations and optimal control

Distinguishing mathematics. The unknown is a curve, function, or control schedule, and stationarity is weaker than global optimality.

Question seeds.

49-S01 For a sufficiently smooth integrand L and C^2 extremizer with fixed endpoints, first variation yields $d/dx L_y = L_y$; this is necessary, not sufficient.

49-S02 The shortest rectifiable Euclidean path between two points is the segment, by the triangle inequality.

49-S03 A frictionless bead released from rest under uniform gravity leads to the brachistochrone time integral; its Euler-Lagrange extremals are cycloidal arcs with appropriate endpoints. [UCL MATH0043, §2](#), “The Brachistochrone”.

49-S04 A small control problem: for $x'=u$, $|u| \leq 1$, moving from 0 to $a>0$ takes at least a time units, achieved by $u=1$ almost everywhere.

Entrances and stops. Design paths of string with specified obstacles; distinguish shortest, fastest, and least bending. Children can compare candidate tracks, but a race is experimental evidence under a physical model. The time-integral investigation begins at calculus; global sufficiency and general control theory require more.

Trap/frontier. Soap films can be trapped in local minima. Euler-Lagrange equations do not certify that any solution minimizes. Relaxation, existence of minimizers, constraints, and Pontryagin's principle require further advanced surveys.

FIELD 51 | 4 SURVEYED SEEDS | GA-21

51 Geometry

Distinguishing mathematics. Axioms, distance, incidence, and transformations generate different geometries; a familiar drawing may support incompatible structures.

Question seeds.

51-S01 For endpoints in the same open half-plane, reflecting one endpoint across a straight mirror converts the shortest one-bounce Euclidean path to a straight segment, provided its crossing lies in the allowed mirror segment.

51-S02 In a Euclidean triangle the angles sum to π ; in a hyperbolic triangle the sum is strictly smaller.

51-S03 Circle inversion sends circles or lines to circles or lines, with a circle through the inversion center becoming a line.

51-S04 A Euclidean composition of two reflections in intersecting lines is a rotation by twice their directed angle. Petrunin, *Euclidean Plane and Its Relatives*, §§5D, 7C, 10B, 11B-C, and 13D.

Entrances and stops. Plan mirror routes, turn folded paper twice, or compare straight travel on several supplied geometric models. Reflection proofs use congruence; inversion needs ratios and coordinates. Hyperbolic-model claims need definitions of distance and straightness, not merely bowed grid lines.

Trap/frontier. Week 9 already uses billiard unfolding; new mirror work must introduce a new constraint or optimization question. Do not confuse projective incidence with Euclidean lengths. Finite geometries, projective duality, constructibility, and buildings warrant separate activity searches.

FIELD 52 | 4 SURVEYED SEEDS | GA-22; GA-31

52 Convex and discrete geometry

Distinguishing mathematics. Global shape and intersection structure can be forced by a few points or small subfamilies.

Question seeds.

52-S01 Any $d+2$ points in $\mathbb{R}^d$ admit a partition into two nonempty parts with intersecting convex hulls (Radon).

52-S02 If every $d+1$ members of a finite family of at least $d+1$ convex sets in $\mathbb{R}^d$ intersect, all members intersect (Helly).

52-S03 Every point in a convex hull in $\mathbb{R}^d$ is a convex combination of at most $d+1$ original points (Carathéodory).

52-S04 Disjoint compact convex sets can be strictly separated by a hyperplane. [Hug and Weil, *A Course on Convex Geometry*, §§1.2 and 1.4; Theorems 1.2.1-1.2.4.](#)

Entrances and stops. Place four pegs in the plane and split them into two colors so their stretched-rubber-band hulls meet. Collinear and interior-point cases must be included. A Helly puzzle can use transparent convex windows; deliberately allow a nonconvex window to test the assumption. Drawing and inclusion suffice to enter; general d -dimensional arguments need linear dependence.

Trap/frontier. Pairwise intersections do not suffice in the plane. Existing tilings cover a different part of discrete geometry. Packing density, polytopes, oriented matroids, rigidity, and geometric discrepancy remain substantial territory.

FIELD 53 | 4 SURVEYED SEEDS | GA-23; GA-30

53 Differential geometry

Distinguishing mathematics. Curvature, intrinsic distance, and transport connect local measurements to global shape without requiring an external view.

Question seeds.

53-S01 A circle of radius r has curvature $1/r$; bending a plane into a cylinder preserves intrinsic distances locally but changes its extrinsic appearance.

53-S02 Nonantipodal points on a round sphere have a unique shortest great-circle arc.

53-S03 A convex geodesic triangle on the unit sphere contained in an open hemisphere, has area equal to its angle sum minus π .

53-S04 Parallel transport around such a triangle rotates a tangent direction by its signed spherical area modulo 2π . Petrunin and Zamora Barrera, *What Is Differential Geometry?*, chapters 14-18, especially 16 and 17C.

Entrances and stops. Carry an arrow north from the equator, turn along a second meridian, and return along the equator; the octant triangle supplies an exact right-angle example. Compare shortest paths on a cylinder by unrolling it. Movement and angle tracking provide access; general curvature and transport equations stop at multivariable calculus.

Trap/frontier. “Walk straight” needs a defined transport rule. Great circles are locally straight, but sufficiently long arcs are not globally shortest. General relativity, symplectic/contact geometry, and comparison geometry require further branches.

FIELD 54 | 4 SURVEYED SEEDS | GA-24; GA-36

54 General topology

Distinguishing mathematics. Continuity, convergence, separation, connectedness, and compactness are studied independently of a particular metric or embedding.

Question seeds.

54-S01 A continuous image of a connected space is connected.

54-S02 A continuous image of a compact space is compact.

54-S03 A continuous bijection from compact X to Hausdorff Y has continuous inverse.

54-S04 The closure of $\{(x, \sin(1/x)) : 0 < x \leq 1\}$ is connected but not path connected. *Sharifi, Point-Set Topology*, chapter 3, especially Proposition 3.1.8 and connected/compact-space sections.

Entrances and stops. Compare a loop and a segment by removing a chosen point and counting pieces; specify the objects as topological spaces, not pencil traces with accidental crossings. A coverage game on $[0, 1]$ can contrast finite subcovers with the family of intervals covering $(0, 1)$ but approaching an omitted endpoint. Elementary examples use continuous deformation and logic; general topology needs quantified definitions.

Trap/frontier. The phrase “rubber-sheet geometry” omits most topology. A continuously traced curve is path connected; that does not characterize connectedness. Finite grids cannot reproduce the sine-curve counterexample. Separation axioms, product topology, dimension, and nonmetrizable spaces remain advanced searches.

FIELD 55 | 4 SURVEYED SEEDS | GA-25

55 Algebraic topology

Distinguishing mathematics. Algebraic invariants turn deformation and obstruction questions into computable evidence.

Question seeds.

55-S01 Loops in a circle based at one point have an integer winding number; concatenation adds it.

55-S02 A connected finite graph with V vertices and E edges has $E-V+1$ independent cycles.

55-S03 The alternating cell count of a finite CW complex equals the alternating ranks of its homology groups.

55-S04 Every continuous map $S^2 \rightarrow \mathbb{R}^2$ identifies an antipodal pair; the $S^1 \rightarrow \mathbb{R}$ case follows from the intermediate value theorem. [Hatcher](#), *Algebraic Topology*, §1.1, Theorem 1.10, §2.2/Theorem 2.44, and §2.B.

Entrances and stops. Wind strings around a forbidden peg with a fixed basepoint, or add corridors to a connected network and predict independent loops. Removing edges of a spanning tree gives a concrete graph proof. The equal-height circle is a genuine continuous entrance. Homology groups and higher-dimensional obstructions need algebra and topology.

Trap/frontier. Counting visible holes is ambiguous unless dimension and space are fixed. Equal Euler characteristic does not imply homeomorphism. Week 1's configuration graph is an overlap, but its connectivity alone does not establish homology of a richer configuration space.

FIELD 57 | 3 SURVEYED SEEDS | GA-26; GA-32

57 Manifolds and cell complexes

Distinguishing mathematics. Local Euclidean structure coexists with global gluing, knots, orientation, and high-dimensional phenomena.

Question seeds.

57-S01 Connected compact surfaces without boundary are classified by orientability and genus: sphere with g handles, or sphere with k crosscaps. [Sahoo](#), *Surfaces*, §8, [classification theorem/corollary](#), pp.17-19.

57-S02 Reidemeister moves preserve the number of Fox 3-colorings of a knot diagram; a nonconstant trefoil coloring distinguishes it from the unknot.

57-S03 A Möbius band has one boundary component and reverses transverse orientation along its central loop; an annulus has two boundary components and preserves orientation. Knot invariance is developed in [Kauffman](#), *Knots*, “Three Colored Trefoil,” [Figures 12-14 and accompanying theorem](#).

Entrances and stops. Glue marked strips, trace edges before cutting, and color knot diagrams while checking local moves. Reading can be minimal; consistency checking is substantive. Classification proofs require gluing words and systematic reduction. Smooth structures, surgery, and high-dimensional manifolds retain university prerequisites.

Trap/frontier. Cutting changes the space and is not a homeomorphism. A physical Klein-bottle model necessarily self-intersects in three-space; that crossing is not intrinsic. Failure of a chosen coloring test does not prove a knot unknotted.

FIELD 58 | 3 SURVEYED SEEDS | GA-27

58 Global analysis, analysis on manifolds

Distinguishing mathematics. Differential equations, critical points, and differential operators interact with the global topology of a manifold.

Question seeds.

58-S01 Near a nondegenerate critical point of a smooth real function, local coordinates make the function a constant plus a sum of positive and negative squares (Morse lemma).

58-S02 A smooth map from a compact manifold slab to $[a,b]$, with boundary at the endpoints and no critical points, has diffeomorphic level surfaces throughout.

58-S03 Crossing one nondegenerate critical point of index k changes the sublevel homotopy type by attaching a k -cell, under compactness and isolated-critical-value hypotheses. Gillespie, *Morse Theory Notes*, Lemma 1.12, Theorem 2.9, §3.

Entrances and stops. Raise an imaginary waterline through a clay landscape, recording component births, merges, and holes. For a deliberately specified smooth surface, this is a model of sublevel topology. Derive and classify critical points of x^2+y^2 and x^2-y^2 with calculus; the general Morse correspondence needs manifolds and topology. A triangulated landscape is a separate piecewise-linear model.

Trap/frontier. A contour drawing does not verify smoothness, nondegeneracy, or absence of hidden critical points. Morse theory is one entrance, not all global analysis. Elliptic operators, heat kernels, index theory, and infinite-dimensional manifolds remain prerequisite-blocked/unresolved.

Editorial decisions and next survey frontier

The most promising new low-prerequisite work is concentrated in **continuous comparison, convexity, transformations, intrinsic geometry, and topological obstruction**. Prioritize the equal-height antipodes, four-point Radon split, one-bounce shortest path with a constrained mirror, cylinder routes, octant parallel transport, convolution overlap, and knot-coloring invariance. Each supplies choices, counterexamples, and a proof question rather than a famous-name demonstration. These are candidates for design/review, not approved worksheets.

A second tranche should retain secondary-school gates: minimax fitting, functional-equation machines over $\mathbb{Q}$ versus $\mathbb{R}$, ODE uniqueness failure, geometric versus harmonic accumulation, finite Fourier reconstruction and aliasing, and separable-kernel integral equations. They extend the library upward by mathematical prerequisites, without forcing calculus vocabulary into younger sessions.

The hard choices were to leave SCV without an elementary bridge; keep spectral and global-analysis theorems behind their real prerequisites; and label graph averaging, finite matrices, and triangulated landscapes according to the structure they actually realize. These restrictions protect the atlas from letting one attractive activity silently “cover” half of analysis.

Current weeks already provide tiling/reconfiguration (Week 1), finite linear transformations (Week 2), compositions (Week 3), discrete orbits (Week 4), and unfolding/periodicity (Week 9), plus networks (Week 10). Those are useful prerequisites and comparison cases. Reuse should introduce an additional invariant, changed space, limiting issue, or genuinely new optimization question; merely changing the story would not expand the child's mathematical experience. The current use log must still distinguish planned from taught instances.

Before leaf-level coverage is claimed, commission protected searches in geometric measure theory, real-variable harmonic analysis, functional/operator algebras, complex geometry, nonlinear PDE, ergodic theory, symplectic/contact geometry, topology beyond surfaces, and index theory. Also inspect the primary source and proof of Poincaré recurrence before turning its screening anchor into a family. This file establishes **27 screened top-level dossiers, 104 anchors, and an explicit frontier**; it establishes no exhaustive leaf audit, independent bridge review, or classroom validation.

PART III

Probability, computation, and applications

Prepared 25 September 2026. Scope: MSC2020 fields **60, 62, 65, 68, 70, 74, 76, 78, 80, 81, 82, 83, 85, 86, 90, 91, 92, 93, 94, 97**: nineteen mathematical content fields and one education/support field. The pinned denominator is taxonomy.json, not a claim that the anchors below exhaust its descendants.

This is research screening with candidate bridges, before independent family review. Each numbered anchor is a distinct question worth designing around. The proposed entrances are our design inferences, not activities validated by their sources. The capability letters follow DESIGN.md; the prose specifies the actual requirement. “Stop” means retain the topic at that prerequisite, not replace its mathematics with a loosely related puzzle. Physical trials provide data; idealized models require separate arguments.

FIELD 60 | 4 SURVEYED SEEDS | AP-01; AP-02

60 Probability theory and stochastic processes

Distinguishing questions. What is predictable about an unpredictable path? How do conditioning, dependence, stopping, rare events, and scaling change an answer? Finite probability is a substantial entrance; it does not cover stochastic calculus or measure-theoretic probability.

60-S01 Conditioning changes the sample space. In a finite model, conditional probabilities are ratios within the event learned. Compare drawing a colored counter, learning that it came from a particular bag, and learning information selected by a biased reporting rule. The information-generating procedure is part of the hypotheses.

60-S02 Gambler's ruin. An independent fair nearest-neighbor walk on integers 0 through N , stopped at an endpoint, hits N with probability i/N from i . Harmonic averaging plus boundary values proves this; biased steps change the answer.

60-S03 Branching and extinction. Independent identically distributed offspring counts give extinction probability equal to the smallest fixed point of their probability generating function on $[0,1]$. Mean offspring alone does not describe every generation; the deterministic one-child case is an important exception to careless critical-extinction slogans.

60-S04 Mixing versus cycling. Finite irreducible aperiodic chains converge to their unique stationary distribution. Alternating deterministically between two states has a stationary distribution but no convergence from a single state.

Entrance/gates. O/C: walk counters and compare endpoint frequencies; F: probability trees and exact proportions; A/P: solve the averaging recurrence; L: transition matrices; D/I/X: diffusion limits, Brownian paths, martingales and stochastic integration. Do not suggest that a fair game guarantees a short play or that a large sample eliminates bias.

Inspected source. Grinstead-Snell, *Introduction to Probability*, CHANCE version, §§4.1, 10.2, 11.4, 12.2; contents and relevant random-walk text inspected. [Author-hosted text](#). These locators are a source queue for the detailed family proof, not a claim that every exercise was checked.

FIELD 62 | 4 SURVEYED SEEDS | AP-03; AP-04; AP-05

62 Statistics

Distinguishing questions. What can incomplete observations establish about an unknown population or mechanism, with what uncertainty and assumptions? This field needs its own designs; calculating dice probabilities is not statistical inference.

62-S01 Sampling distributions and estimator choice. Repeated random samples produce different estimates. In independent sampling, an average can be unbiased while its uncertainty remains appreciable; a convenient or selectively observed sample need not be unbiased.

62-S02 Randomization tests. Under a sharp no-treatment-effect null and the actual randomized assignment mechanism, enumerate treatment-label assignments and compare a prespecified statistic. The exact reference distribution comes from the design, not from “everything seems random.”

62-S03 Interval coverage. A procedure’s confidence level concerns its repeated-sampling coverage. Bootstrap samples draw with replacement from the empirical distribution; a bootstrap interval is not automatically exact, especially with tiny or unrepresentative samples.

62-S04 Association, confounding, prediction. Compare a pooled relationship with the within-group relationships, or compare training fit with held-out prediction. More variables or a higher-degree curve can improve fit without improving prediction or supporting a causal conclusion.

Entrance/gates. O/C: secretly fill jars and let groups choose how to sample; record distributions of guesses. F: compare proportions and averages. A/P: define statistic, null, and assignment space; simulation can precede calculus. L/D: regression geometry and likelihood optimization. A low-threshold task must preserve an unknown target, observed evidence, and justified uncertainty. Avoid interpreting a p-value as the probability the null is true, or showing only a single attractive sample.

Inspected sources. Allen Downey, *Think Stats*, ch. 8, Estimation, ch. 9, Hypothesis Testing; OpenIntro, *Introduction to Modern Statistics*, Foundations of inference, chapters 11-13. Exact finite designs are preferred for the first cards; population claims remain conditional on sampling.

FIELD 65 | 3 SURVEYED SEEDS | AP-06; AP-07

65 Numerical analysis

Distinguishing questions. When an exact formula is unavailable, how can an approximation be computed, certified, and kept stable? Distinguish sensitivity of the mathematical problem from error introduced by the algorithm.

65-S01 Bracketing versus tangent iteration. Continuity and opposite endpoint signs guarantee a bisection root bracket; each step halves its width. Newton iteration additionally needs derivatives and suitable starting conditions for convergence. For $f(x)=x^3-2x+2$, starting at zero produces $0 \rightarrow 1 \rightarrow 0$ exactly.

65-S02 Conditioning versus residual. A small equation residual need not mean a small solution error. In $f(x)=\varepsilon x-1$ with ε positive and tiny, changing the right side slightly produces a large absolute change in x ; scaling and relative error must be stated.

65-S03 Time stepping can invent instability. Euler's update for $y'=-ay$, $a>0$, is $y[n+1]=(1-ah)y[n]$. Decay in magnitude requires $0<ah<2$, although the continuous solution always decays. Halving steps is meaningful only alongside an error/stability question.

Entrance/gates. O/V: repeatedly trap a balance point between two marks; F/A: halving intervals and recurrence tables; D: tangents, Taylor error, ODE interpretation; L: ill-conditioned systems; X: functional error norms for PDE methods. A ruler gives measurement uncertainty, not an exact real oracle. An Euler table is an exact recurrence and an approximation to the ODE, with separate claims.

Inspected sources. Yano-Penn-Konidakis-Patera, *Math, Numerics, and Programming*, Unit VI, §29.2, especially §29.2.5; Driscoll-Braun, *Fundamentals of Numerical Computation*, chapters 1, 6 and 11 located by contents. The three elementary claims above also admit direct algebraic checks.

FIELD 68 | 4 SURVEYED SEEDS | AP-08

68 Computer science

Distinguishing questions. What can a finite rule system recognize or compute, and how do its time, memory, randomness, and interaction requirements grow?

68-S01 Finite memory and distinguishable histories. A deterministic finite automaton recognizing strings whose number of red symbols is divisible by three needs three distinguishable remainder states. A diagram is an exact machine once alphabet, start state, transitions, and acceptance are specified.

68-S02 Recognition versus generation. Balanced parentheses need unbounded counting depth in the unrestricted problem; a fixed maximum depth admits a finite-state machine. “Our sample machine works” does not settle all lengths.

68-S03 Checking versus finding. A satisfying assignment to a finite Boolean formula can be checked efficiently. Encoding a family of puzzles as satisfiability is a reduction only after proving both directions and bounding encoding size; a difficult classroom puzzle does not establish NP-hardness.

68-S04 Diagonal limits. The halting problem for arbitrary programs is undecidable. A finite collection of small programs can be exhaustively analyzed, so finite testing does not instantiate the unrestricted impossibility theorem.

Entrance/gates. O/C: walk through labeled machine states and invent accepted/rejected strings; A/P: reason over arbitrary lengths, construct a separating suffix; X: program encodings, self-reference and reductions. Sorting/search can be a second strand, but should not displace semantics, formal languages, verification, learning, or complexity from the frontier.

Inspected source. Michael Sipser, MIT [18.404J lecture collection](#), lectures 1-5, 8-9, 14-16, and [lecture 1 slides](#). Course structure inspected; full proofs beyond the small automata remain a family-stage obligation.

FIELD 70 | 3 SURVEYED SEEDS | AP-09

70 Mechanics of particles and systems

Distinguishing questions. How do forces, constraints, symmetry and conserved quantities determine motion? A mechanism should include a time evolution, not just a static balance picture.

70-S01 Symmetry and conservation. If a Lagrangian has no dependence on a coordinate, its conjugate momentum is conserved along Euler-Lagrange solutions. Energy conservation similarly requires time-translation symmetry with the appropriate hypotheses.

70-S02 Normal modes. Linearization about a stable equilibrium reduces coupled small oscillations to an eigenvalue problem. Two identical masses coupled by springs have collective in-phase and out-of-phase motions.

70-S03 Stationary action. For smooth fixed-endpoint variations, stationary action produces the Euler-Lagrange equations. The physical path need not globally minimize action; “nature always takes the shortest path” is false.

Entrance/gates. O/V: move a linkage, count independently adjustable joints, compare coupled pendulum motions; this is experimental modeling. F/A: ideal collision momentum and energy bookkeeping; L: modes; D/I: action and nonlinear motion. Conservation can be explored below calculus using exact collision rules, while proving the general symmetry theorem stops at calculus. Friction, external forces, and large amplitudes challenge the idealization productively.

Inspected source. David Tong, *Classical Dynamics*, §§2.1, 2.3-2.6, particularly the stationary-versus-minimum warning, constrained coordinates, two-body reduction and small oscillations. Hamiltonian recurrence and chaos remain separate follow-up territory.

FIELD 74 | 3 SURVEYED SEEDS | AP-10

74 Mechanics of deformable solids

Distinguishing questions. How does a body's shape change under loading, and when does it buckle, yield, or fracture? Constitutive assumptions are mathematical input, not incidental engineering detail.

74-S01 Series versus parallel elasticity. Ideal Hookean springs in series add compliances; parallel springs add stiffnesses. Equal displacement and equal force are different constraints. This is an exact small network of constitutive laws.

74-S02 Stress, strain and anisotropy. Force is not stress, and extension is not strain. Two geometrically similar specimens with different cross-sections need different loads for the same stress; direction-dependent material response is distinct from geometry-dependent stiffness.

74-S03 Buckling as loss of stability. An ideal pinned slender Euler-Bernoulli column first buckles at $P = \pi^2 EI / L^2$ under its model assumptions. Different end conditions alter the answer; an imperfect strip can bend before the ideal threshold.

Entrance/gates. O/V: compare folded paper strips and elastic arrangements, choose a fair comparison; F: force per area and relative extension; A: spring equations; D/I/L: beam equations and eigenmodes. A spring network is a finite model, not proof of continuum elasticity. Do not claim that all rubber bands are Hookean, or that stiffness and strength are synonymous.

Inspected source. Allan Bower, *Applied Mechanics of Solids*, §3.2, especially 3.2.1-3.2.3; contents, chapters 5, 9, 10 locate energy, failure, rods and beams. Buckling formula is a candidate requiring its beam chapter checked before a full family is accepted.

FIELD 76 | 3 SURVEYED SEEDS | AP-11

76 Fluid mechanics

Distinguishing questions. How can local conservation laws describe deforming continuous matter, and when can viscosity, compressibility, rotation or turbulence be neglected?

76-S01 Conservation along a stream tube. In steady incompressible flow without leaks, volume flux is constant, so mean speed rises when cross-sectional area decreases. Unsteady storage invalidates that shortcut.

76-S02 Bernoulli's relation. For steady inviscid flow with conservative body force, the energy expression is constant along a streamline; equality across different streamlines needs additional assumptions. Faster flow and lower pressure is not a universal law for every experiment.

76-S03 Low-Reynolds-number reversibility. Negligible inertia gives linear Stokes equations. Reciprocal back-and-forth shape changes cannot propel the standard isolated swimmer under the scallop-theorem assumptions. Boundaries, non-Newtonian fluids or extra degrees of freedom change the setting.

Entrance/gates. O/V: move material strips to distinguish pathlines from a snapshot of arrows; F: flow-rate conservation; A/V: pressure-energy bookkeeping; D/I/X: vector calculus, PDEs, Reynolds number and viscous asymptotics. Beads on a graph can model a conservation network but omit deformation and shear. The swimmer is promising for a visual adolescent investigation, not yet a verified low-preparation child activity.

Inspected source. Tong, *Fluid Mechanics*, [chapter 1: inviscid flows](#), continuity and Euler/Bernoulli development; [course structure](#) locates viscous flow and instabilities. The scallop anchor needs an additional primary source before promotion beyond candidate status.

FIELD 78 | 3 SURVEYED SEEDS | AP-12; AP-13

78 Optics, electromagnetic theory

Distinguishing questions. How do fields superpose and propagate, and how do geometry, material response and boundary conditions shape waves?

78-S01 Reflection and refraction. Reflecting one endpoint converts a shortest broken path at a flat mirror into a straight-line problem. With different constant speeds in two media, minimizing travel time leads to Snell's law; minimum distance gives the wrong refracted path.

78-S02 Electrostatic boundary problems. Potentials superpose in a linear fixed-medium model, but conductor boundaries constrain the solution. Discrete harmonic averaging has a maximum principle; its grid model must be distinguished from the continuum Laplace equation.

78-S03 Interference and polarization. Coherent waves add amplitudes, not intensities. Two equal opposite-phase waves cancel in the ideal linear model. Projections through ideal polarizers depend on angle; adding a middle polarizer can transmit light through an otherwise crossed pair.

Entrance/gates. O/V: mirror paths and translucent wave strips; F/A: weighted travel times and signed amplitudes; trigonometry: refraction and polarization; D/I/X: Maxwell equations, boundary-value PDEs and diffraction integrals. String or water waves share superposition but are not electromagnetic fields. A laser drawing alone does not prove Fermat's principle.

Inspected source. Tong, *Electromagnetism*, course chapters 1-3, electrostatic potentials, wave propagation and polarization; *Electrodynamics* chapter, §§4.3.2-4.3.3, PDF pp. 20-24, polarization and reflection, inspected after the older PDF link failed. Refraction and the three-polarizer claim still need their exact source checked at family stage.

FIELD 80 | 3 SURVEYED SEEDS | AP-14

80 Classical thermodynamics, heat transfer

Distinguishing questions. What transformations are possible under energy and entropy constraints, and how does heat move through space? Keep thermodynamic state functions separate from microscopic explanations in field 82.

80-S01 Path dependence versus state. Internal energy is a state function, whereas heat and work depend on a process. On a pressure-volume cycle, signed enclosed area is work for the quasistatic idealization.

80-S02 Carnot bound. A heat engine between reservoirs at absolute temperatures $T_h > T_c$ has efficiency at most $1 - T_c/T_h$; reversible engines attain the ideal bound. Celsius ratios and hidden extra reservoirs invalidate the usual calculation.

80-S03 Diffusion and maximum principles. Source-free heat diffusion smooths temperature differences under appropriate boundary conditions. An insulated body's total heat is conserved; fixed-temperature boundaries exchange heat, so the boundary specification matters.

Entrance/gates. O/F: exchange measured warm/cool water and compare equilibrium predictions under equal specific-heat assumptions; V/A: area and energy ledgers; D/I: diffusion equations and entropy integrals. Averaging counters is an exact discrete process, a shared mechanism with diffusion, and not a derivation of the heat equation. Stop before claiming entropy is simply “messiness.”

Inspected source. Tong, *Statistical Physics*, ch. 4, *Classical Thermodynamics*, laws, heat engines and thermodynamic potentials. Heat-PDE family verification should additionally use the numerical-analysis/analysis dossier; it is not supplied by thermodynamics alone.

FIELD 81 | 3 SURVEYED SEEDS | AP-15

81 Quantum theory

Distinguishing questions. What changes when states are amplitudes in a Hilbert space and observables are noncommuting operators? The probabilistic output does not make quantum mechanics ordinary hidden dice.

81-S01 Amplitude interference. Adding state amplitudes and then squaring magnitudes differs from adding classical probabilities. Two real two-component vectors already provide a meaningful restricted calculation; complex relative phases extend it.

81-S02 Measurement order. Projective measurements in different bases generally do not commute. Specify normalized state, projectors, outcome probabilities and state update; a decorative “quantum coin” omitting these is insufficient.

81-S03 Discrete spectra from boundaries. A particle in an infinite one-dimensional well has standing-wave eigenstates with energies proportional to n^2 . The boundary conditions, Hamiltonian and normalization are essential, not arbitrary numbering of allowed shapes.

Entrance/gates. V/F: projections and squared lengths for a real-amplitude special case; L/A: matrices, norms and probabilities; complex arithmetic: general finite states; D/I/X: Schrödinger equations and unbounded operators. A three-polarizer exploration is an analogue until the single-photon measurement model is specified. Finite-state quantum examples can stop below calculus while the well genuinely requires it.

Inspected source. Tong, *Quantum Mechanics*, ch. 3 §§3.1-3.4, and [chapter map including the one-dimensional well](#). Entanglement, quantum computation and scattering remain distinct frontier mechanisms.

FIELD 82 | 3 SURVEYED SEEDS | AP-16

82 Statistical mechanics, structure of matter

Distinguishing questions. How do microscopic states and interactions produce macroscopic behavior, including sharp phase changes and fluctuations?

82-S01 Multiplicity versus a macrostate. Counting microstates with a fixed energy or particle total explains why some macrostates are more numerous. Equal likelihood of microstates is an assumption to inspect, not a consequence of counting.

82-S02 Competing energy and entropy. A finite Ising model assigns spins and interaction energy; canonical weights are proportional to $\exp(-E/kT)$. Enumerating a four-site system exposes alignment versus multiplicity without claiming a true finite-temperature singular phase transition.

82-S03 Thermodynamic limits and correlations. Large-system limits can be nonanalytic even when every finite partition function is smooth. Mean-field self-consistency and exact low-dimensional models answer different questions and need different qualifications.

Entrance/gates. O/C: list two-color spin arrangements, count agreeing neighbors, compare energy ties; F: weighted sampling; A: exponentials and partition functions; D/I/X: infinite-volume limits, criticality and correlation functions. Ordinary majority voting shares an interaction motif but is not the canonical Ising measure. A tiny model honestly covers a Hamiltonian and ensemble, not the general phase-transition theorem.

Inspected sources. Tong, *Statistical Physics*, ch. 1, [ensemble foundations](#), and [chapter 5 outline](#) for Ising, mean field and criticality. Small finite partition functions are ideal candidates for exact independent enumeration.

FIELD 83 | 3 SURVEYED SEEDS | AP-17

83 Relativity and gravitational theory

Distinguishing questions. How do invariant spacetime intervals, causal structure and curvature govern observation and motion? Ordinary curved-surface geometry is useful preparation but does not itself represent Lorentzian spacetime.

83-S01 Light cones and invariant interval. In 1+1 special relativity, $c^2\Delta t^2 - \Delta x^2$ is invariant under Lorentz transformations. Events may be timelike, null or spacelike separated; arbitrary rotations of a Euclidean drawing do not preserve this interval.

83-S02 Proper time along paths. Different timelike worldlines between fixed events can accumulate different proper times. Piecewise inertial paths offer a calculable twin-clock example with square roots and specified turnaround.

83-S03 Geodesics in curved spacetime. A metric determines a connection and test-particle geodesics; the Einstein equations additionally determine the metric from matter. Solving motion in a supplied metric is not solving the gravitational field equations.

Entrance/gates. V/A: space-time coordinates, speeds and square roots; D/I: proper-time integration; L/X: tensors and differential geometry for GR. A light-clock diagram can give a faithful special-relativistic argument. A rubber-sheet model is motivation only and can misleadingly import an external downward force. It is acceptable for most of this field to stop at algebraic high-school or calculus access.

Inspected source. Tong, *General Relativity*, ch. 1 §§1.1-1.3, particularly equations (1.8), (1.28)-(1.35), parameter constraints and the Schwarzschild example.

FIELD 85 | 3 SURVEYED SEEDS | AP-18

85 Astronomy and astrophysics

Distinguishing questions. How can motion and radiation constrain systems that cannot be directly manipulated, and how do gravitating systems form or remain stable?

85-S01 Orbit inversion. In a two-body inverse-square model, period and orbital scale constrain mass; an observed projection does not reveal every orbital parameter. Conics are consequences of the force law, not merely shapes to draw.

85-S02 Virial balance. For a suitable bounded, time-averaged gravitational system, $2 \langle T \rangle + \langle U \rangle = 0$. This connects velocity dispersion and size to mass; it is not an instantaneous identity for every collapsing cloud.

85-S03 Jeans instability. Linearized homogeneous self-gravitating gas balances pressure against gravity; wavelengths beyond its Jeans scale grow in the ideal model. The model's background assumptions and linearization must be explicit.

Entrance/gates. V/F: compare orbit traces and inverse-square scaling with geometric measurements; A: infer a parameter under an assumed model; D//L: orbit ODEs, virial derivation and perturbation eigenmodes. Accessible data puzzles should preserve uncertainty and model dependence. “Make constellations” has no substantive bridge to these mechanisms. Real astronomy may require a supplied data set and facilitator preparation beyond counters.

Inspected source. MIT, *8.902 Astrophysics II* (2023), [full lecture notes](#), PDF pp. 15-16, virial theorem assumptions; PDF pp. 45-46, Jeans scale and crossing/free-fall times. Orbit work also connects to Tong's mechanics, §2.5.4. Stellar structure and radiative transfer remain explicit gaps.

FIELD 86 | 4 SURVEYED SEEDS | AP-19

86 Geophysics

Distinguishing questions. How do rotation, stratification, waves and limited observations constrain Earth's fluids and interior? Shared PDEs do not erase the field's distinctive balances and inverse problems.

86-S01 Rotating-frame balance. In a rapidly rotating, slowly evolving large-scale flow, pressure-gradient and Coriolis terms can approximately balance, giving geostrophic flow along pressure contours. Water simply flowing downhill is a misleading default.

86-S02 Potential-vorticity conservation. In an ideal inviscid shallow-water layer, (relative vorticity + Coriolis parameter)/depth is materially conserved. Changing depth or latitude links column stretching and rotation; forcing and friction break the conservation statement.

86-S03 Wave-speed inverse problems. In a layered model with specified ray assumptions, different travel times can reveal interfaces or speeds, but nonuniqueness and incomplete observations remain. Finite ray sums are an honest algebraic special case of a larger inverse problem.

86-S04 Depth controls long-wave speed. Small-amplitude, nonrotating shallow-water gravity waves have speed $\sqrt{gH}$ under the long-wavelength approximation. This predicts changes in travel time without equating the wave's motion with the motion of an individual water parcel.

Entrance/gates. V/F: follow tracer paths or infer two unknown travel speeds from a few measured times; A/L: solve inverse systems and test whether data suffice; D/I/X: rotating shallow-water PDEs and continuous tomography. A turntable demonstration is an experiment, not a miniature ocean with matched dimensionless parameters. Seismology and geomagnetism require additional sources before claiming subfield coverage.

Inspected source. Geoffrey Vallis, *Atmospheric and Oceanic Fluid Dynamics*, [author-hosted excerpts](#): contents locate §§3.4, 3.6.1 and 5.1 for geostrophy and potential vorticity; §7.1, equations (7.18)–(7.23), PDF pp. 75–76, inspected for the shallow-water limit and parcel motion. The [author's edition page](#) distinguishes sample from full book. This supports the fluid strand, not the proposed seismological extension.

FIELD 90 | 4 SURVEYED SEEDS | AP-20; AP-21

90 Operations research, mathematical programming

Distinguishing questions. How can competing feasible decisions be optimized, and how can one certify that no better decision exists? Include continuous convexity, uncertainty and scheduling, not only shortest paths.

90-S01 Convex local-to-global reasoning. For a convex objective on a convex feasible set, every local minimum is global. Strict convexity ensures at most one minimizer, but neither existence nor feasibility follows automatically.

90-S02 Dual bounds as certificates. A feasible primal maximization allocation and feasible dual resource prices give lower and upper bounds; equality certifies optimality. Strong duality needs its own hypotheses outside ordinary feasible finite linear programs.

90-S03 Dynamic programming. A finite-horizon state captures everything needed for the remaining cost; solve backward by comparing immediate cost plus optimal future cost. A greedy next move can fail when it ignores later resource consequences.

90-S04 Integer constraints matter. A fractional relaxation gives a bound but may not yield an implementable integer allocation. Rounding can violate resource constraints; integrality is a theorem in particular structures, not a general convenience.

Entrance/gates. O/C: build products from limited colored pieces, choose bundles, devise a price-based impossibility certificate; F/A: continuous allocations and inequalities; P: justify a bound for all choices; L/D: duality and convex optimization. A finite toy factory is exact integer optimization; its continuous relaxation is a second model.

Inspected source. Boyd-Vandenberghe, *Convex Optimization*, chapters 2-5, especially local/global optimality and weak/strong duality. Dynamic programming and discrete optimization deserve separate source checks at family stage rather than being attributed to this convex text.

FIELD 91 | 3 SURVEYED SEEDS | AP-22; AP-23

91 Game theory, economics, social and behavioral sciences

Distinguishing questions. What changes when outcomes depend on other agents' choices, preferences and information? Optimization for one planner and equilibrium among several participants are different questions.

91-S01 Mixed strategies. A finite two-player zero-sum game has a minimax value when randomized strategies are allowed. Matching pennies has no pure-strategy equilibrium; making the randomizer observable changes the game.

91-S02 Nash equilibrium versus social optimum. A finite game has a mixed equilibrium, but equilibrium can be inefficient and need not be unique. In a small congestion game, compare unilateral incentives with total cost.

91-S03 Coalitions and allocation. A transferable-utility coalition game specifies what each coalition can obtain. A core allocation must resist every coalition; the core can be empty. A proposed notion of fairness is an axiom choice, not a mathematical universal.

Entrance/gates. O/C: play a simultaneous-choice matrix game with cards; F: compare expected payoffs; A/P: solve indifference equations and test every profitable deviation; X: fixed-point existence arguments. Nim from prior weeks is a valid combinatorial-game branch but should not stand in for strategic or economic games. Children's observed play is not proof of equilibrium or of rational-behavior assumptions.

Inspected sources. Thomas Ferguson, [authored game-theory course and notes outline](#), Part II §§1-4, Part III §§1-2, Part IV §§1-3. Old linked text paths fail, so this supplies scope evidence only. Giacomo Bonanno, [Game Theory](#), §§1.6, 5.2-5.3, Theorems 5.1-5.2 and the mixed-strategy calculations on printed pp. 195-199, supplies inspected strategic-game exposition. Coalitional claims still require a detailed source or a self-contained finite proof.

FIELD 92 | 4 SURVEYED SEEDS | AP-24; AP-25

92 Biology and other natural sciences

Distinguishing questions. Which mathematical mechanisms explain growth, competition, transmission, heredity and reaction, and when do competing models make distinguishable predictions?

92-S01 Logistic growth versus exponential growth. $dN/dt=rN(1-N/K)$, $r,K>0$, has equilibria 0 and K with different stability. A chosen discrete update can overshoot or become negative; it is not automatically the same population model.

92-S02 Epidemic thresholds. In a closed homogeneous-mixing SIR model, infections initially grow when $\beta S(0)>\gamma$ under the convention $S'=-\beta SI$. Susceptible depletion can stop growth without every individual being infected.

92-S03 Genetic drift versus selection. Randomly sampling the next finite generation can eliminate an allele even when its expected fraction is unchanged. Selection, mutation and population size are separate parameters, not explanations to interchange casually.

92-S04 Stoichiometry and reaction rates. A reaction network has conserved linear combinations of species counts; mass-action rates then determine dynamics. Atom bookkeeping alone does not establish a reaction's speed or equilibrium.

Entrance/gates. O/C: draw next-generation colored counters or move counters between S/I/R compartments under fully specified rules; F: proportions and repeated sampling; A: recurrences and conservation; L/D: stability, kinetics and population equations. Token processes are exact stochastic/discrete models, not automatically numerical solutions of a deterministic ODE. Avoid implying that a model is a forecast without parameter estimation and empirical checks.

Inspected source. Jeffrey Chasnov, *Mathematical Biology*, §§1.1-1.4, 4.1-4.5, 5.5, 6.1-6.2; preface and chapter structure inspected. Mathematical ecology, biochemical reaction systems and sequence alignment receive separate branches; Fibonacci alone would underrepresent this field severely.

FIELD 93 | 3 SURVEYED SEEDS | AP-26; AP-27

93 Systems theory; control

Distinguishing questions. What can a system's inputs change, what can its outputs reveal, and can feedback achieve a goal despite disturbance and imperfect knowledge?

93-S01 Feedback gain and stability. In $x[n+1]=ax[n]+u[n]$, choosing $u[n]=-kx[n]$ produces multiplier $a-k$. Convergence to zero requires $|a-k|<1$. Stronger correction can create oscillation or divergence.

93-S02 Reachability. For a finite-dimensional linear time-invariant system, the columns $B, AB, \dots, A^{n-1}B$ determine its reachable directions. Missing directions cannot be repaired merely by trying more controls.

93-S03 Observability. Distinct internal states can generate the same measured output sequence. A sensor that reads the sum of two identical independent modes may fail to reveal their difference; adding a second sensor can remove that ambiguity.

Entrance/gates. O/V: steer a paper position marker under a fixed update rule, choose corrections, compare delayed information; F/A: signed multipliers and recurrence behavior; L: rank, modes and observability; D/I: continuous feedback and optimal control. Finite reachability games are exact control systems only when dynamics, input restrictions and observations are specified. A feedback experiment by itself does not cover robust control or Kalman filtering.

Inspected source. Åström-Murray, *Feedback Systems*, [chapter 7](#), [State Feedback](#), [summary items 1-6](#), and [chapter 8 scope](#). The scalar discrete example is directly proved by its geometric progression. Advanced rank and estimation claims remain explicit proof gates.

FIELD 94 | 4 SURVEYED SEEDS | AP-28; AP-29; AP-30

94 Information and communication, circuits

Distinguishing questions. What resources are required to encode, convey, reconstruct or protect information through constrained or noisy channels?

94-S01 Variable-length coding. Unequal message probabilities can make unequal codeword lengths efficient, but unique decoding matters. Prefix-free binary codewords form leaves of a binary tree; allowing a codeword to prefix another can introduce ambiguity.

94-S02 Error detection versus correction. Minimum Hamming distance d detects every error pattern of fewer than d changed bits and corrects every pattern of at most $\lfloor (d-1)/2 \rfloor$ errors by nearest codeword. The assumed error model is essential.

94-S03 Capacity versus finite performance. Shannon's noisy-channel theorem concerns asymptotic reliable transmission below capacity under its channel assumptions; a particular short repetition code does not reach the theorem's guarantee.

94-S04 Circuit laws and transients. Kirchhoff conservation plus component laws give a solvable linear resistive network; capacitors introduce state and RC transients. This deserves a separate circuit strand rather than being swallowed by communication games.

Entrance/gates. O/C: send counter patterns through a partner who flips a specified number; choose redundant encodings; F: weighted average length; A/L: linear codes or circuits; D/I: signal filtering and transients; X: asymptotic coding theorems. A secret cipher is not automatically error correction. Current Week 6 query-code work is prior-use adjacency, not a new family merely by renaming the story.

Inspected source. Claude Shannon, *A Mathematical Theory of Communication*, Parts I-II, source coding example on PDF p. 18 and noiseless/noisy coding statements. Hamming distance and circuits need their own detailed sources or self-contained proofs in later cards; Shannon's paper is not a universal citation for all field 94.

FIELD 97 | 3 SURVEYED SEEDS | support dossier

97 Mathematics education: research-support disposition

Distinguishing questions. Which representations support learners' mathematical decisions? What changes between entering, exploring, explaining and proving? What evidence separates an attractive proposal from a successful teaching design?

97-S01 Representation and transfer. Compare the same mathematical relation represented by objects, diagrams and symbols. Similar surface performance need not mean that a child recognizes the shared structure.

97-S02 Participation and proof. Record independent examples, counterexamples and explanations, alongside hints and adult assistance. Solving quickly and understanding generally are separate observations.

97-S03 Design and assessment. Revise a task in response to observed obstacles, preserving version and context. A repeated success with one group supports local feasibility; it does not justify universal grade labels or causal effects.

Source distinction. Rozhkovskaya's *Math Circles for Elementary School Students*, local EPUB OEBPS/part0010.xhtml, "Introduction: Berkeley 2009," explicitly reports grades 1-3 groups following the same lesson plans and the aim of individual adult assistance; part0027.xhtml, "Introduction: Manhattan 2011," reports locally adapted organization and community support. These are authored practice accounts, not controlled evidence that our proposed variants will work. Our suggestions about logging and comparison are inferences.

Disposition. This field supports design, review and pilot protocols; it does not need a student worksheet labeled "mathematics education." Gates concern facilitator observation and educational research methods. Reading, working memory, motor access and mathematical prerequisites must be recorded separately. The local source was actually inspected; avoid inventing child responses or classroom results.

Protected frontier and handoff

The first design batch should contain genuinely different mechanisms: a random walk; a sampling or randomization inference; a numerical failure; an automaton; coupled motion; constitutive constraints; continuous flow; wave interference; thermodynamic process; finite quantum measurement; finite statistical ensemble; spacetime geometry; astronomical inference; rotating-fluid or inverse-data model; dual optimization certificate; strategic equilibrium; stochastic population model; feedback/observability; error-correcting communication. Several require adolescent algebra, calculus or linear algebra. That is successful coverage of prerequisites, not a failure to reach elementary school.

Important remaining territories include stochastic calculus and large deviations; causal identification and Bayesian decision theory; PDE numerical convergence; machine learning and formal verification; rigid-body chaos; nonlinear elasticity and fracture; turbulence and multiphase fluids; diffraction and inverse scattering; irreversible thermodynamics; quantum many-body theory; renormalization; Einstein field equations; stellar structure and radiative transfer; solid-Earth inversion; stochastic/integer optimization; mechanism design and social choice; evolution, reaction-diffusion and sequence comparison; robust/optimal control; cryptography, continuous signals and circuit synthesis. These are **uninvestigated descendants or source-located candidates**, not implicitly covered by the initial anchors.

All external URLs were checked in this research pass. A successful source lookup or contents read is distinguished above from inspection of the relevant exposition. Failed old links are recorded rather than substituted with uninspected mirrors. No sources were downloaded into this file's working area, no pilot evidence was generated, and no PDFs were created.

The next research frontier

An atlas can have a complete *index* without having completed the mathematics. This edition pins all 6,603 rows of the official downloaded taxonomy. The top-level survey is a breadth pass. The 534 subarea rows, 5,503 ordinary subject rows, and 503 cross-cutting rows remain individually unvisited in the coverage ledger until a later record justifies another status.

The next unit of research

Take a subarea, read an authored overview and at least one precise problem/theorem source, and record 3-5 *distinct questions*, not synonyms for the field. For each question, try a finite object, lower dimension, restricted parameters, geometric model, part of a proof, or genuine symbolic investigation. Preserve failed bridges and the reason they fail. A failed search is useful, but a quick failure is not a theorem that no accessible entrance exists.

Use these dispositions per explicitly investigated anchor: candidate; exact special case; faithful representation; shared mechanism; motivation only; prerequisite stop; unresolved; duplicate with a reason. A field label and a source citation are not themselves a disposition for its descendants.

Protected searches after the breadth edition

Territory	Specific work still needed	Why a small existing example does not finish it
Logic and foundations	Model theory, proof theory, computability, set theory, independence	Truth tables do not instantiate independence from an axiom system.
Number theory	Analytic distribution, algebraic integers, local fields, Diophantine geometry	Modular clocks do not cover analytic number theory or arithmetic geometry.
Abstract algebra	Modules, representation theory, homological constructions, noncommutative phenomena	A group action or matrix example has a limited anchor.
Algebraic geometry and K-theory	Schemes, sheaves, intersection theory, vector bundles and higher invariants	A polynomial plot or formal difference is an entrance to one construction only.
Continuous analysis	Measure, convergence modes, differentiation, singularities, function spaces	Finite counting, numerical samples and smooth examples miss pathologies.
Geometry and topology	Curvature, Riemann surfaces, manifolds, homotopy, homology, characteristic classes	A cut-and-glue surface does not establish the classification theorem.
Differential equations	Existence/uniqueness, weak solutions, shocks, stability, boundary conditions	A recurrence model has to earn any claimed continuum limit.
Harmonic and spectral analysis	Approximation, Fourier convergence, uncertainty, infinite spectra	Decomposing four numbers does not prove Fourier-series convergence.
Probability and statistics	Conditional models, dependence, estimation, identifiability, experimental design, causal assumptions	Dice experiments do not by themselves justify an inference.
Algorithms and complexity	Reductions, undecidability, randomized algorithms, approximation, lower bounds	An inefficient puzzle solution does not prove a complexity lower bound.
Applied mathematics	Model derivation, dimensions, conservation, inverse problems, parameter estimation	A visually similar toy is not automatically the physical equation.
Mathematical physics	Relativity, quantum theory, statistical mechanics and fields	Classical probabilities are not quantum amplitudes; a rubber sheet is not spacetime.
Interfaces	Tropical geometry, optimal transport, topological data, information geometry, machine learning theory	Intersections deserve searches, not only multiple labels on one card.

From families to a six-year circle

The elementary goal remains twenty sessions a year for six years. Ninety atlas families across all prerequisite levels would **not** equal ninety elementary sessions, and a hundred parameter changes would not automatically equal a hundred worthwhile investigations. Build six-year routes only from activities whose gates fit the actual children, allowing genuinely different revisits and several sessions for rich families.

For a proposed return, record the previous exact instance and what changes: the object, allowed action, target question, proof, or connection. A changed context alone does not reset prior use. Retain a reserve of unused instances, and record what children actually did rather than marking the whole packet 'covered'.

The next production selection should deliberately include: a non-counting concrete investigation, a continuous one, a statistical inference question, a symbolic algebra question, and one genuinely calculus-gated investigation. That selection tests the atlas across its intended range. Classroom feedback then informs which other families deserve detailed student pages.

Review limitations

Computational checks certify only their explicitly enumerated domains. A rigorous general argument is recorded separately. Agent review is mathematical/editorial review; it is not a human elementary educator's assessment or a classroom pilot. Source access failures remain visible rather than being converted into verified citations. No current research problem is declared 'still open' merely because it appears in an older source.